Physiology of pain in physical therapy



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Topics covered in the lecture

Neuroanatomy and definition of pain? (International definition of pain)

Dual nature of pain: fast pain and slow pain

What causes pain : pain stimuli

Nerve pathways carrying pain signals to the brain

Brain areas involved in pain perception

Pain modulatory pathways

Neurochemicals involved in pain pathways

Gate control theory of pain







What is pain?

Patients use different words to describe pain

eg.
Aching, Pins and needles, Annoying, Pricking, Biting, Hurting, Radiating, Blunt, Intermittent, Burning, Sore, Miserable, Splitting, Cutting, Nagging, Stabbing, Crawling, Stinging, Crushing, Tender, Dragging, Numbness, Throbbing, Dull, Overwhelming, Tingling, Electric-shock like, Penetrating, Tiring, Excruciating, Piercing, Unbearable

Definition of pain?

There is an International definition of pain formulated by the IASP (International Association for the study of pain)

Pain is an unpleasant sensory and emotional experience associated with actual or potential tissue damage or described in terms of such damage.



Let us think about pain




Let’s start with acute pain



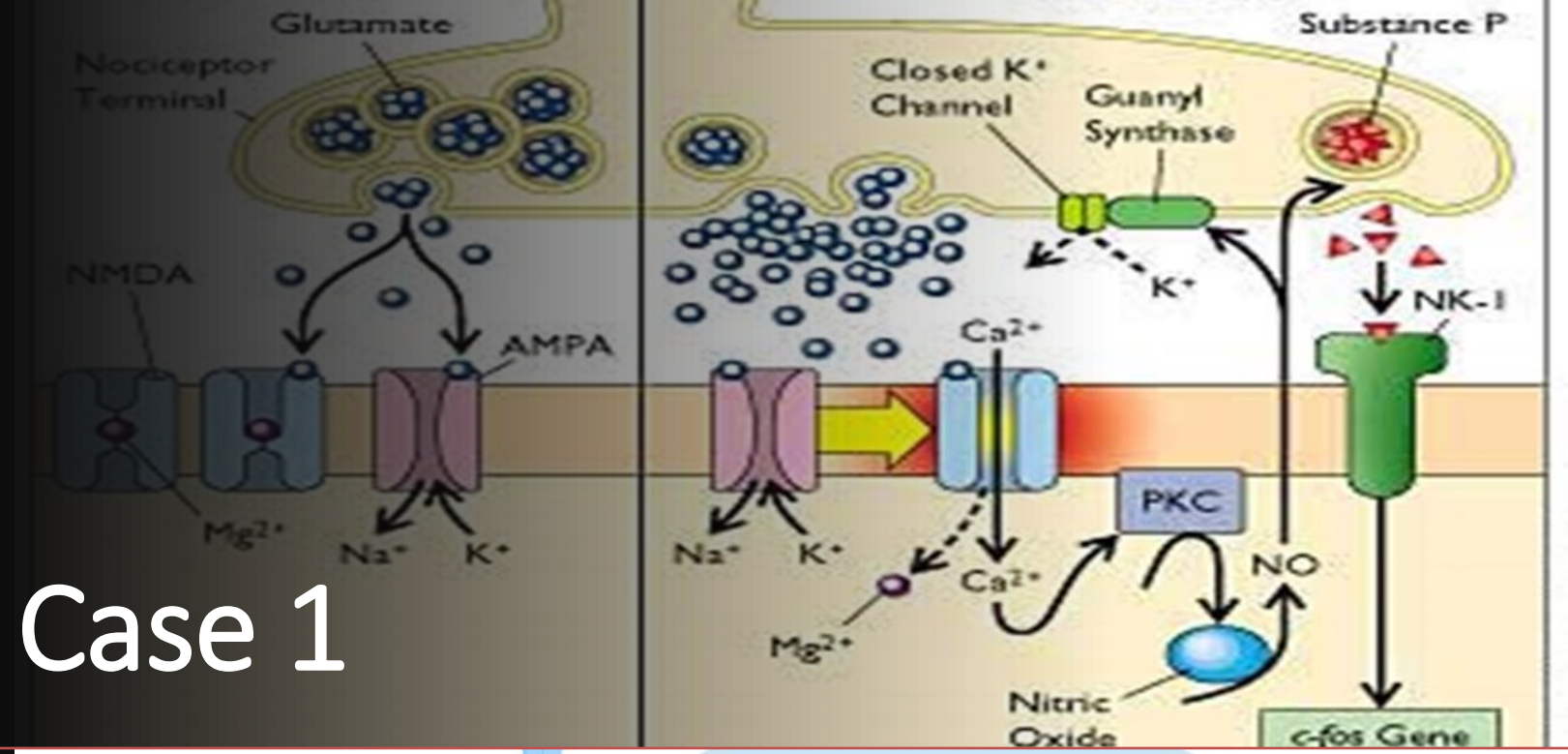
Acute pain

- Tissue damage
- Site and intensity correlation
- Gets better with healing (self limiting)

Case 1

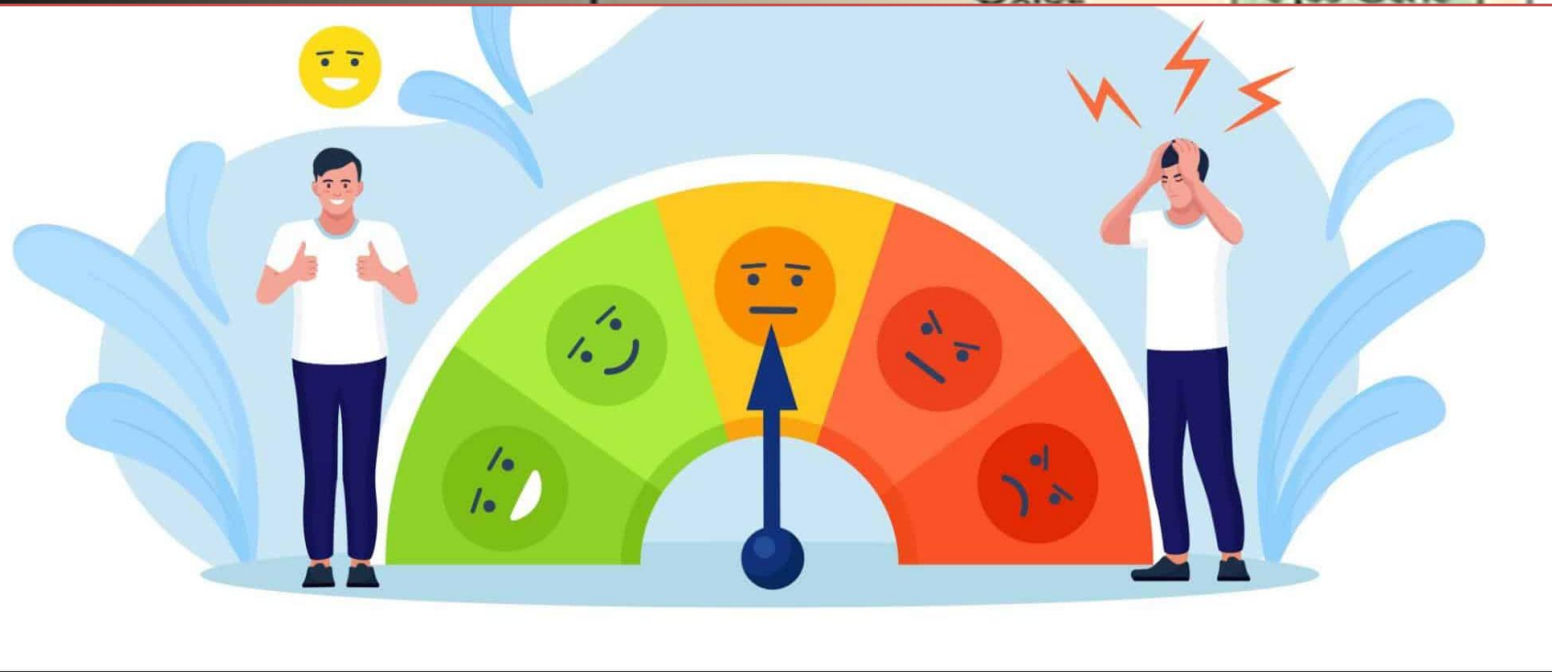
- A man with a pain in his right leg
- “Are you sure it is the ***right*** leg?”
- A man with a pain in his right leg
- How does it feel like?
- any other abnormalities?
- What causes it?





Remarks from Case 1

Chronic pain is ***not*** prolonged acute pain



Remarks from Case 1

- Pathophysiology is ***different*** from acute pain
- Pathophysiology is different from acute pain
- Neuropathic pain
- Site
- Character
- Timing
- More than that…
- Sensitization
- Reduced pain threshold (hyperalgesia)
- Non-painful stimulus (allodynia)



What is pain?

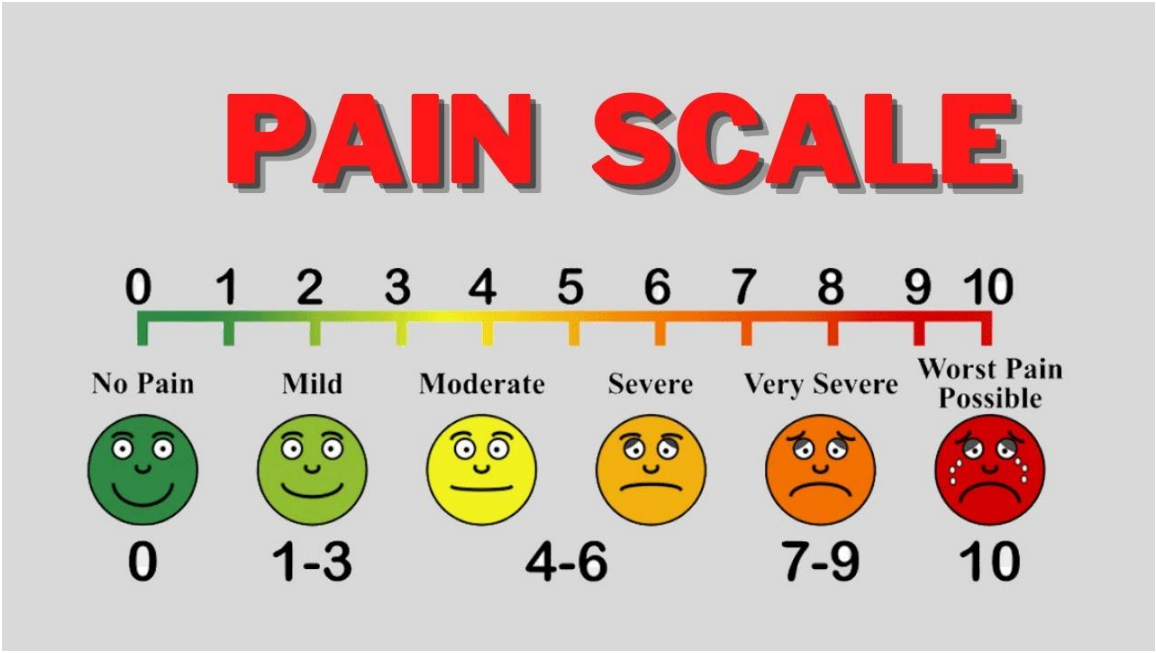


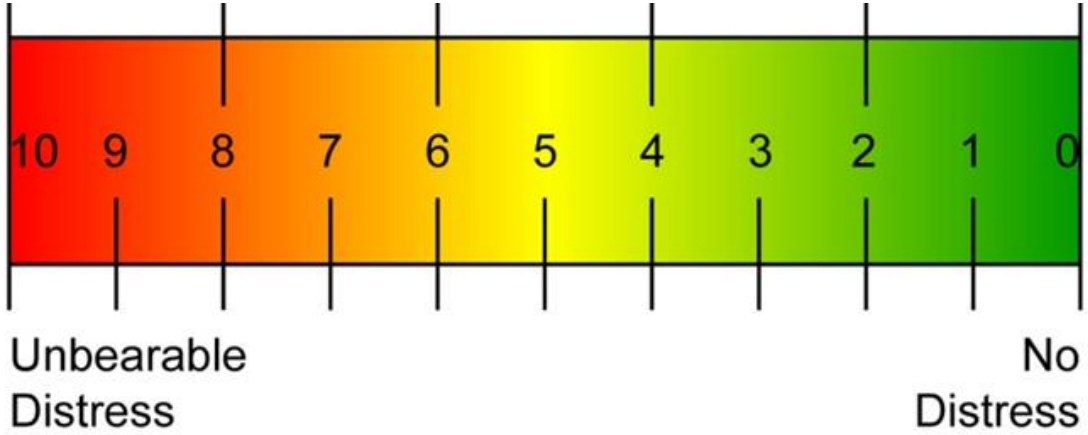
pain

- Pain is
  - subjective
  - protective
  - and it is modified by developmental, behavioural, personality and cultural factors
- It is a symptom
- Associated signs are crying, sweating, increased heart rate, blood pressure, behavioural changes etc









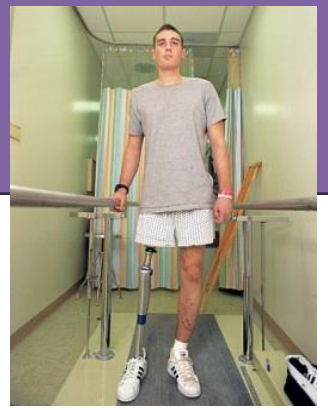
Measurement of pain

- It is difficult to describe pain although we know what it is
- It is difficult to measure pain
  - visual analogue scale (VAS) is used

Dual nature of pain

- Fast pain
  - acute
  - pricking type
  - well localised
  - short duration
  - Thin myelinated nerve fibres are involved (A delta)
- Slow pain
  - chronic
  - throbbing type
  - poorly localised
  - long duration
  - Unmyelinated nerve fibres are involved (c fibres)

Different situations

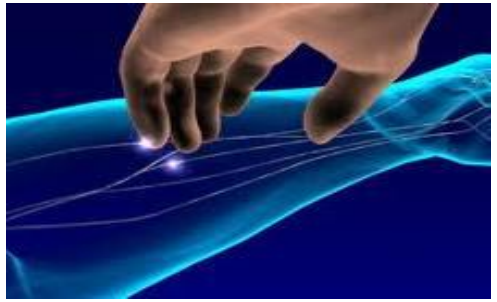


- **No stimuli, but pain is felt**
  - phantom limb pain
    - eg. in amputated limb
    - **Stimuli present, but no pain felt**
      - eg. soldier in battlefield, sportsman in arena





- **Pain due to a stimulus that does not normally provoke pain**
  - Allodynia



- **Pain caused by a lesion or disease** of the somatosensory nervous system
  - Neuropathic pain



Pain terminology

International Association for the Study of Pain 2011

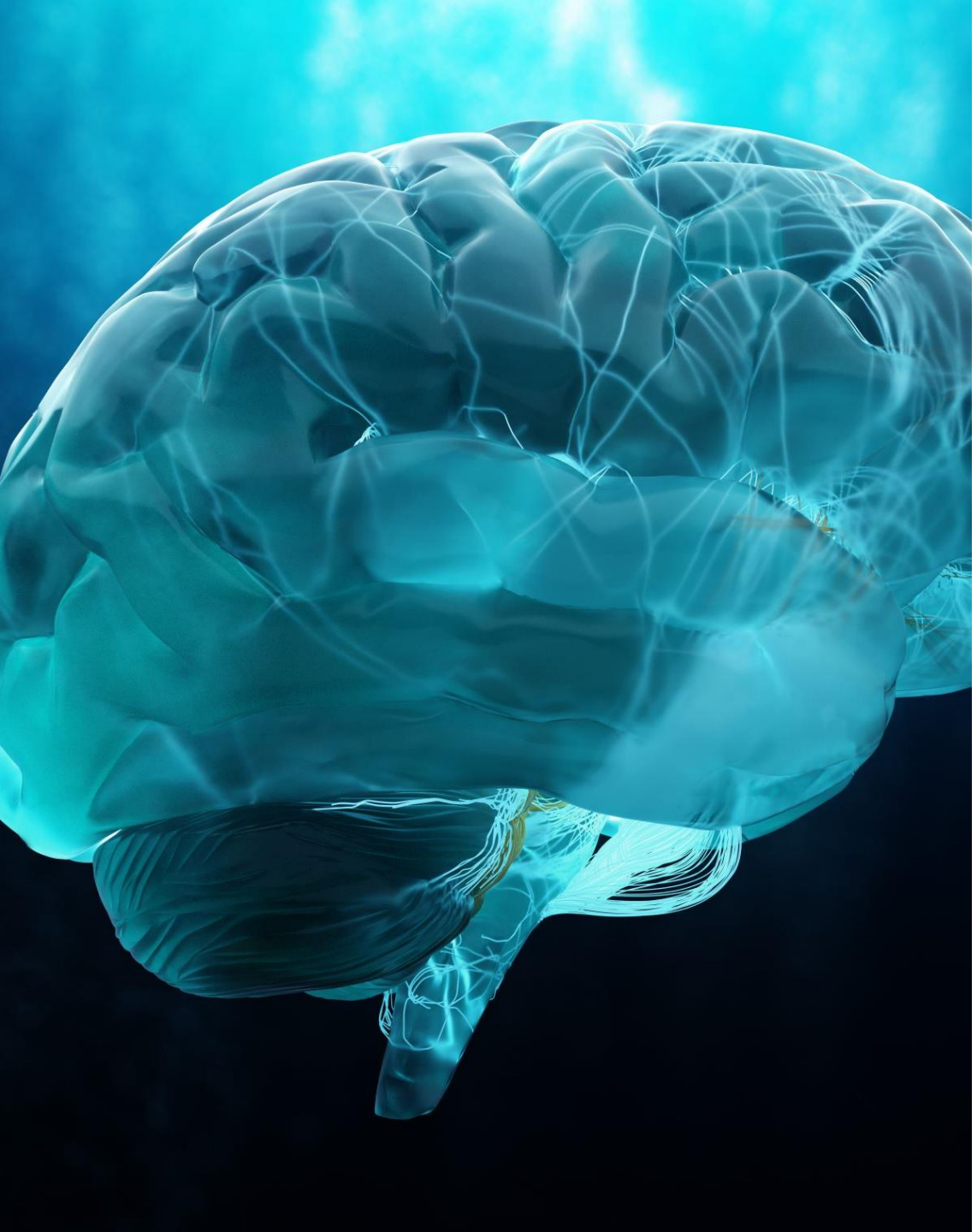
- **Hyperalgesia**
  - *Increased pain from a stimulus that normally provokes pain*
- **Hyperaesthesia**
  - *Increased sensitivity to stimulation, excluding the special senses (increased cutaneous sensibility to thermal sensation without pain)*
- **Paraesthesia**
  - *An abnormal sensation, whether spontaneous or evoked*



Neuralgia	Analgesia	Allodynia	Anaesthesia
Pain in the distribution of a nerve or nerves	Absence of pain in response to a normally painful stimulus	Pain due to a stimulus that does not normally provoke pain	A loss of sensation resulting from pharmacologic depression of nerve function or from neurological dysfunction

Pain

- **Pain as a sensation**
  - physiologically (nociception)
  - Nociceptive pain
- **Pain as an emotional experience**
  - Psychologically
  - Psychogenic pain
- **Pain caused by damage to nerve**
  - Neuropathic pain

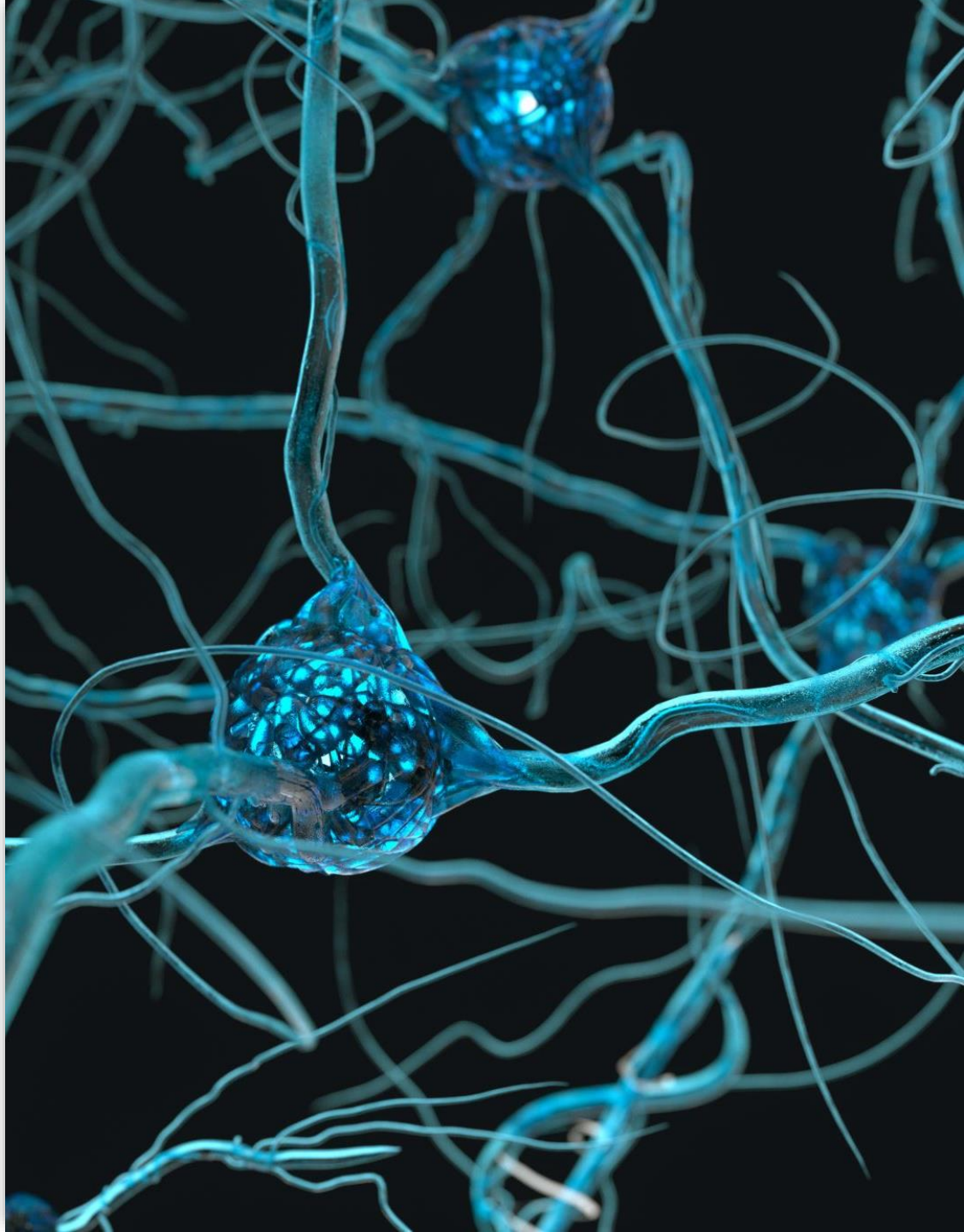


Transduction and perception

- **Transduction**
  - **Transduction** is the process that converts a sensory signal to an electrical signal to be processed in a specialized area in the brain.
  - This takes place at the sensory receptor. The change in electrical potential that is produced is called the receptor potential.

Perception

- Perception is an individual's interpretation of a sensation. Although perception relies on the activation of sensory receptors, **perception happens, not at the level of the sensory receptor, but at the brain level.** The brain distinguishes sensory stimuli through a sensory pathway: action potentials from sensory receptors travel along neurons that are dedicated to a particular stimulus.

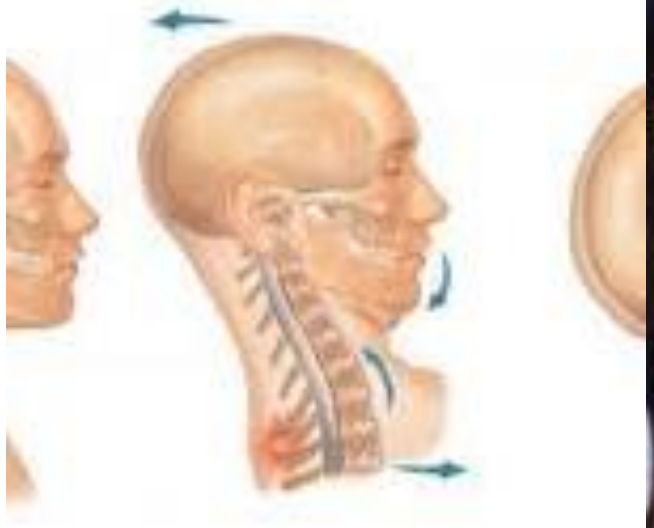


Stimuli

- **Physical**
  - **pressure etc**
- **Electrical**
- **Thermal**
  - **cold, hot**
- **Chemical**
  - **H+, lactic acid, K+, histamine, bradykinin, serotonin, acetylcholine, proteolytic enzymes, capsiacin**
  - **Prostaglandins (PGE2)**
    - **Cannot directly stimulate nociceptors**
    - **Increase the sensitivity of nociceptors for other stimuli (decrease the threshold)**









CLASSIFICATION OF PAIN

PAIN

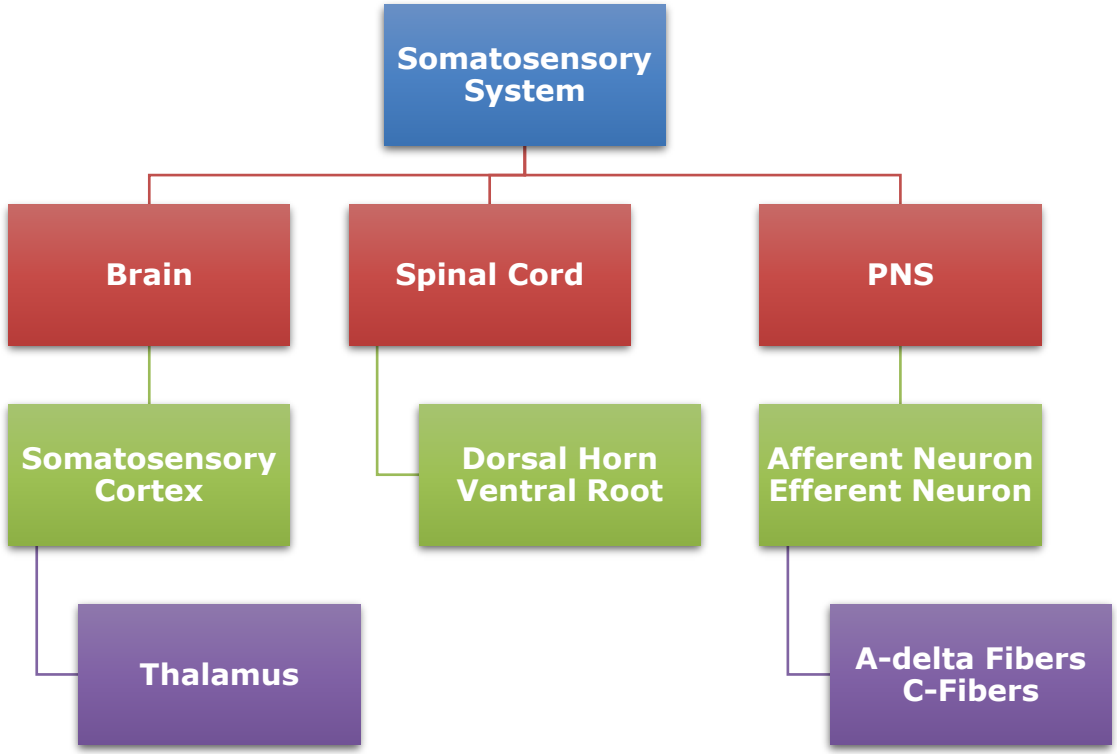
Somatic (somasthetic)

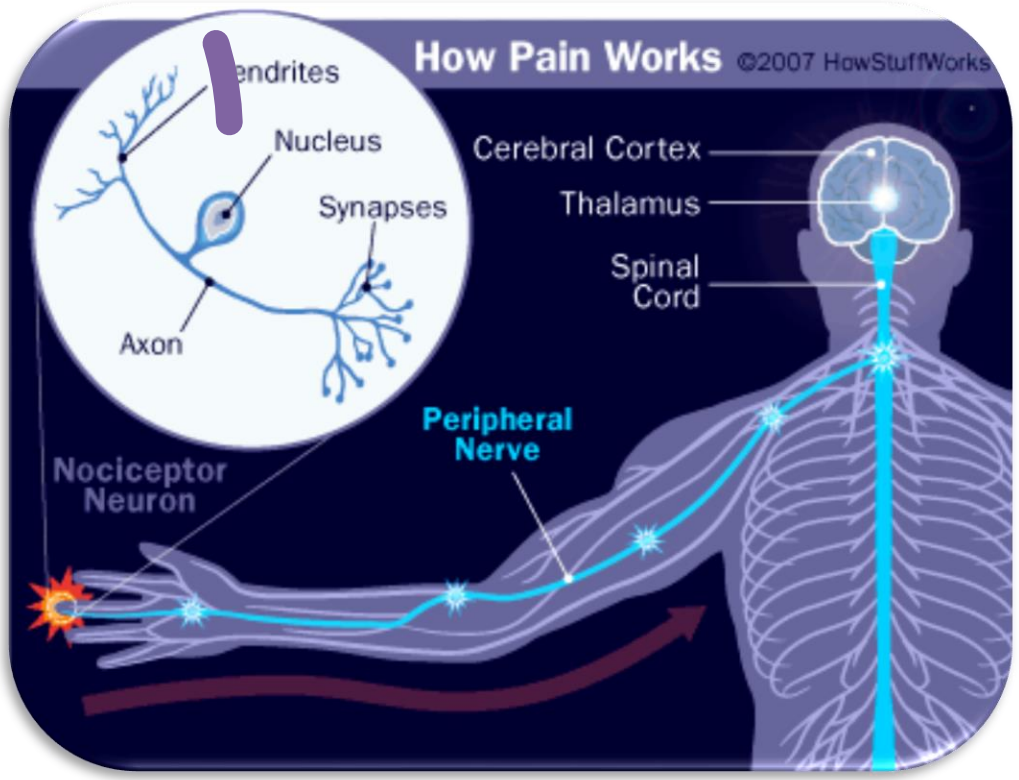
Visceral (from viscera) e.g. angina pectoris, peptic ulcer, intestinal colic, renal colic, etc.

Superficial (from skin & subcutaneous tissue) e.g. superficial cuts/burns, etc.

Deep (from muscles/bones/fascia/periosteum) e.g. fractures/arthritis/fibrositis, rupture of muscle belly

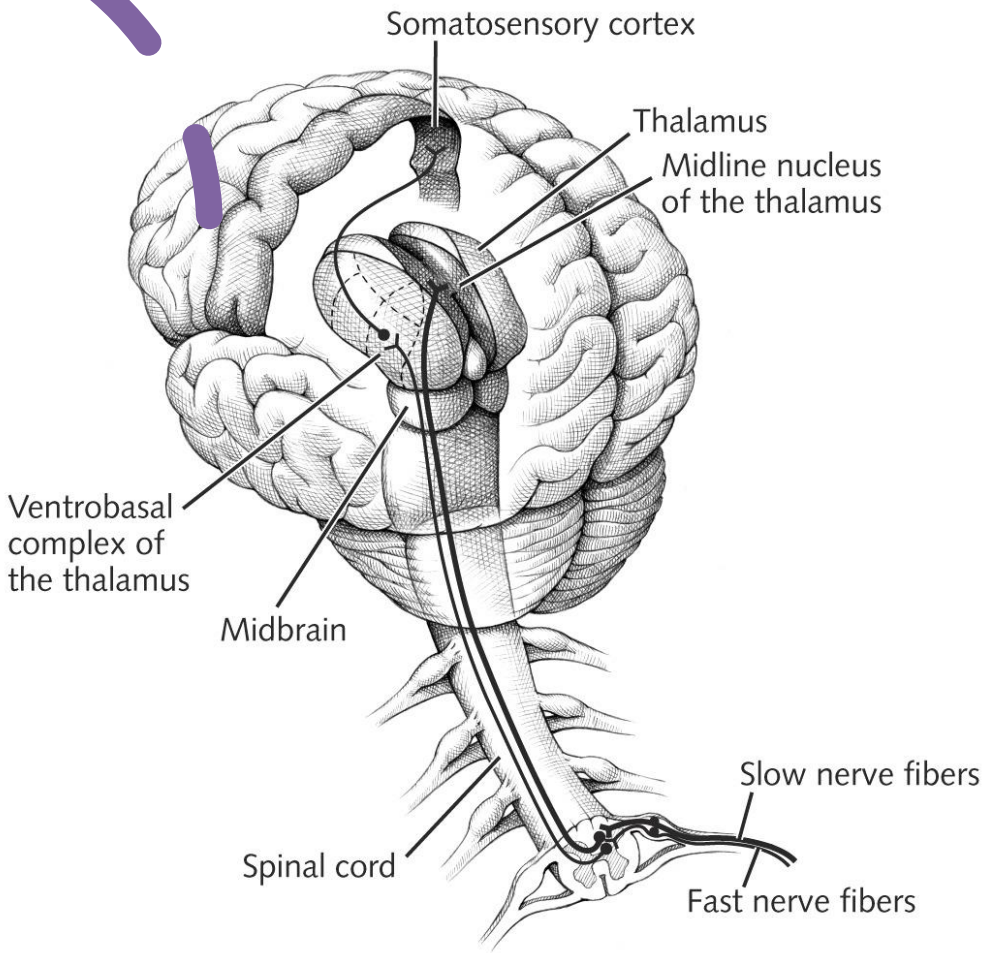
The Nervous System and Pain

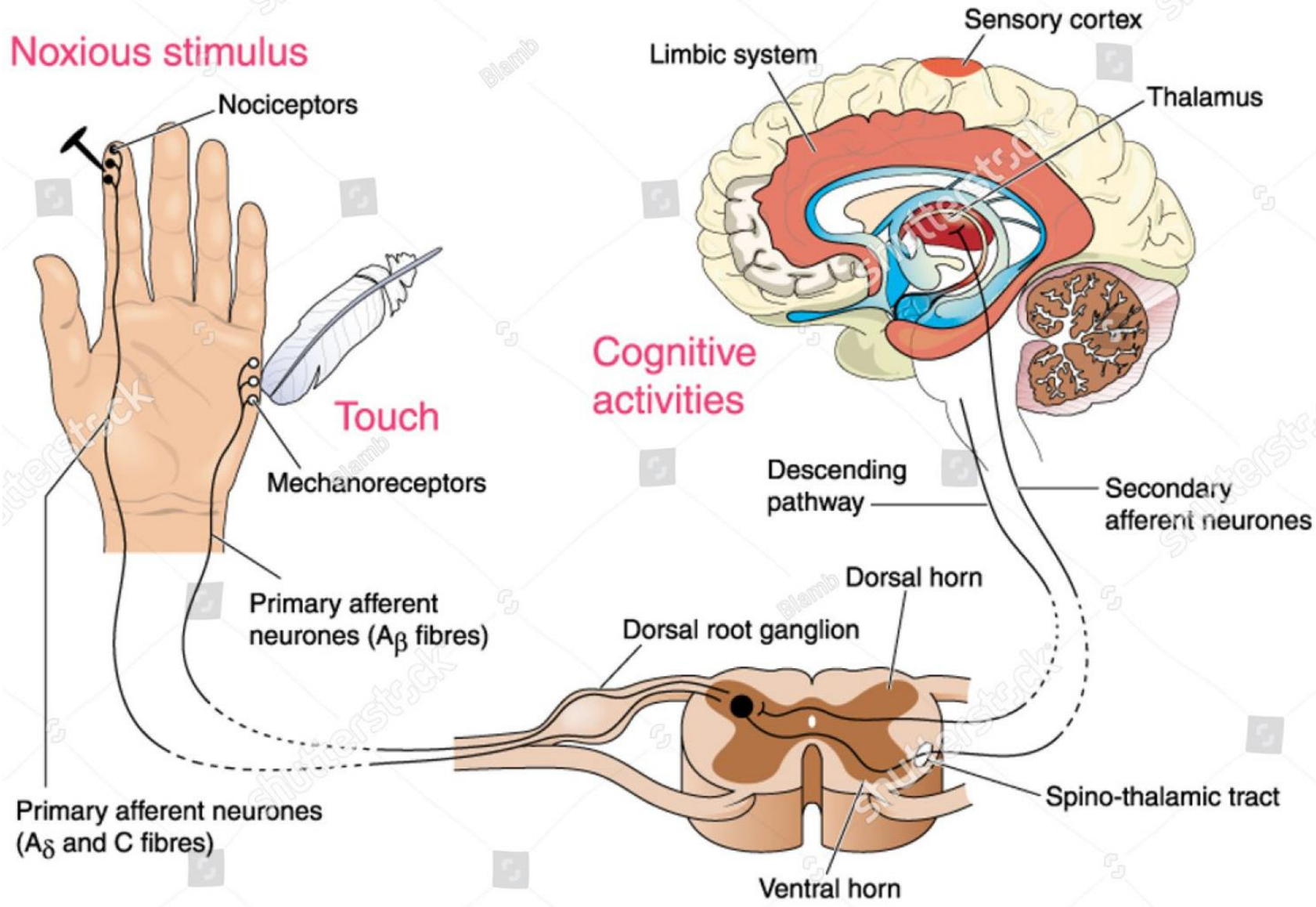




Pain Pathways – Going Up

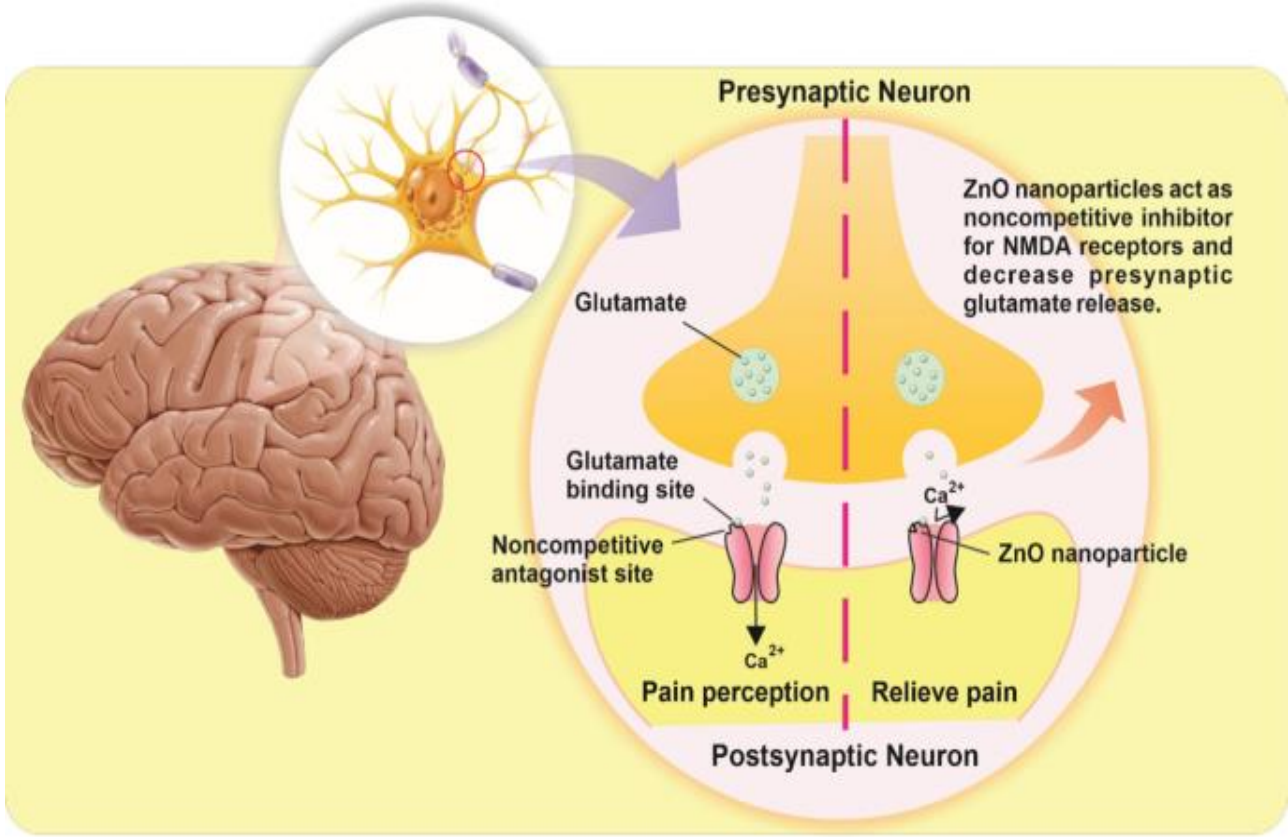
- Pain information travels up the spinal cord through the spino-thalamic track (2 parts)
    - Immediate warning of the presence, location, and intensity of an injury
    - Slow, aching reminder that tissue damage has occurred




Noxious stimulus
Nociceptors
Touch
Mechanoreceptors
Primary afferent neurones (Aβ fibres)
Primary afferent neurones (Aδ and C fibres)
Sensory cortex
Limbic system
Thalamus
Cognitive activities
Descending pathway
Secondary afferent neurones
Dorsal horn
Dorsal root ganglion
Spino-thalamic tract
Ventral horn

The Neurochemicals of Pain

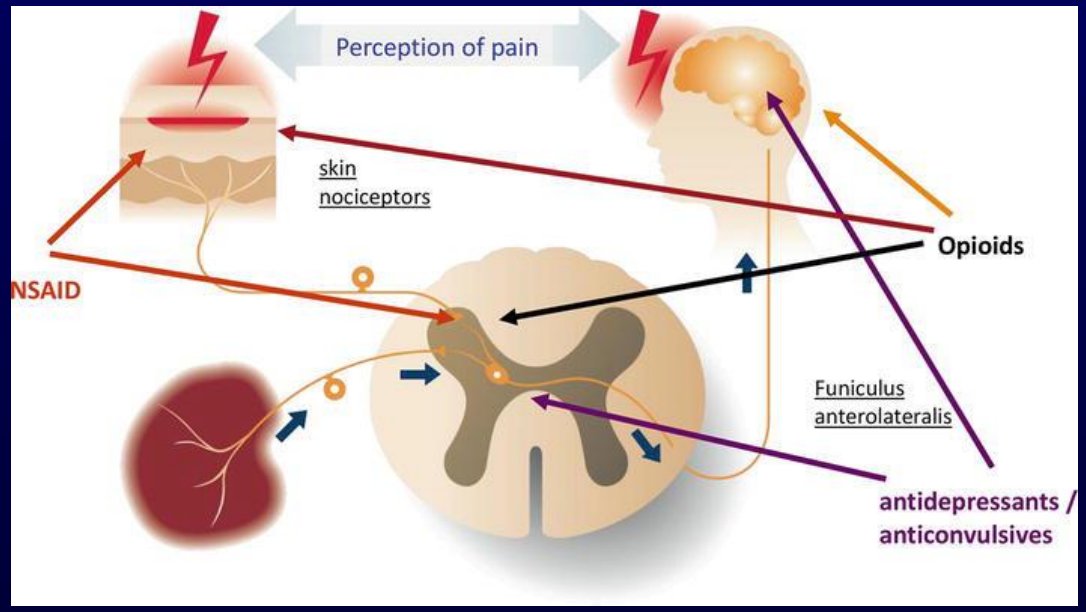
- **Pain Initiators**
  - **Glutamate – Central (Acute – A delta)**
  - **Substance P – Central (chronic – c fiber)**
  - **Brandykinin - Peripheral**
  - **Prostaglandins - Peripheral**
- **Pain Inhibitors**
  - **Serotonin**
  - **Endorphins**
  - **Enkephalins**
  - **Dynorphin**




PRACTICE
MAKES
PERFECT

Define each of the following

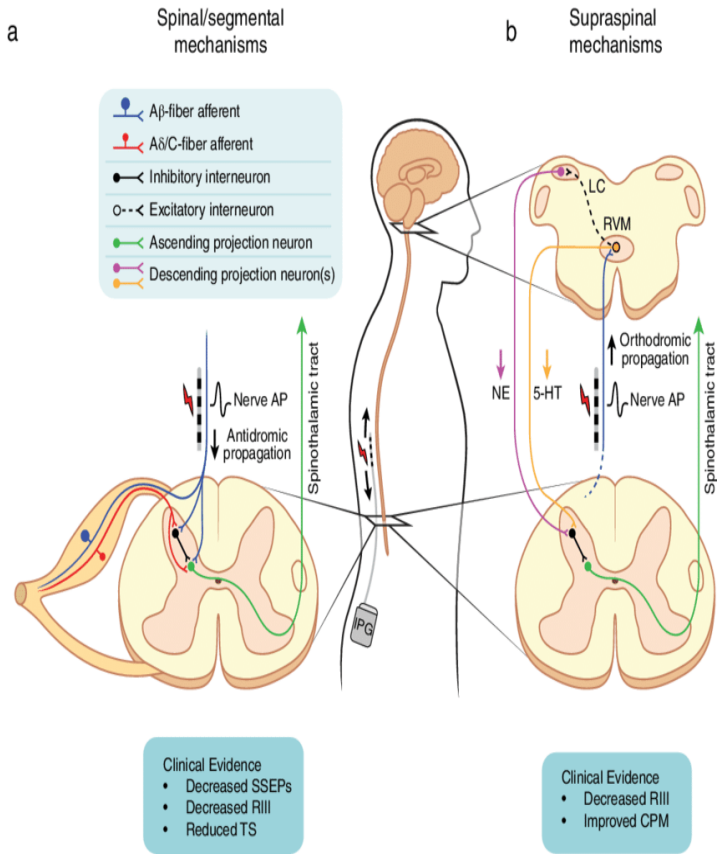
1. Transduction and perception.
2. Enumerate **Pain Inhibitors**
3. Enumerate **Pain Initiators**
4. **Allodynia.**
5. Dual nature of pain
6. **How to measure pain?**



I Sensory pathways



- Sensory systems allow us to detect, analyze and respond to our environment
- “ascending pathways”
- Carry information from sensory receptors to the brain.
- Conscious: reach cerebral cortex.
- Unconscious: do not reach cerebral cortex.
- Sensations from body reach the opposite side of the brain



1. Sensory receptors

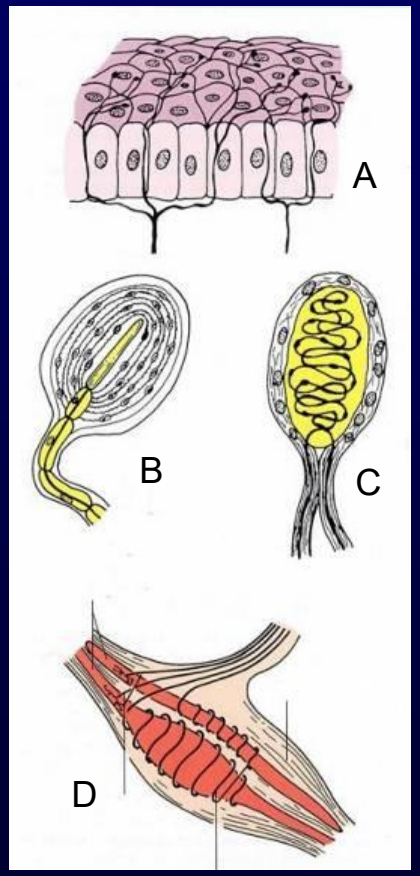
A: Free nerve endings (pain, temperature)

B: Pacinian corpuscle (pressure)

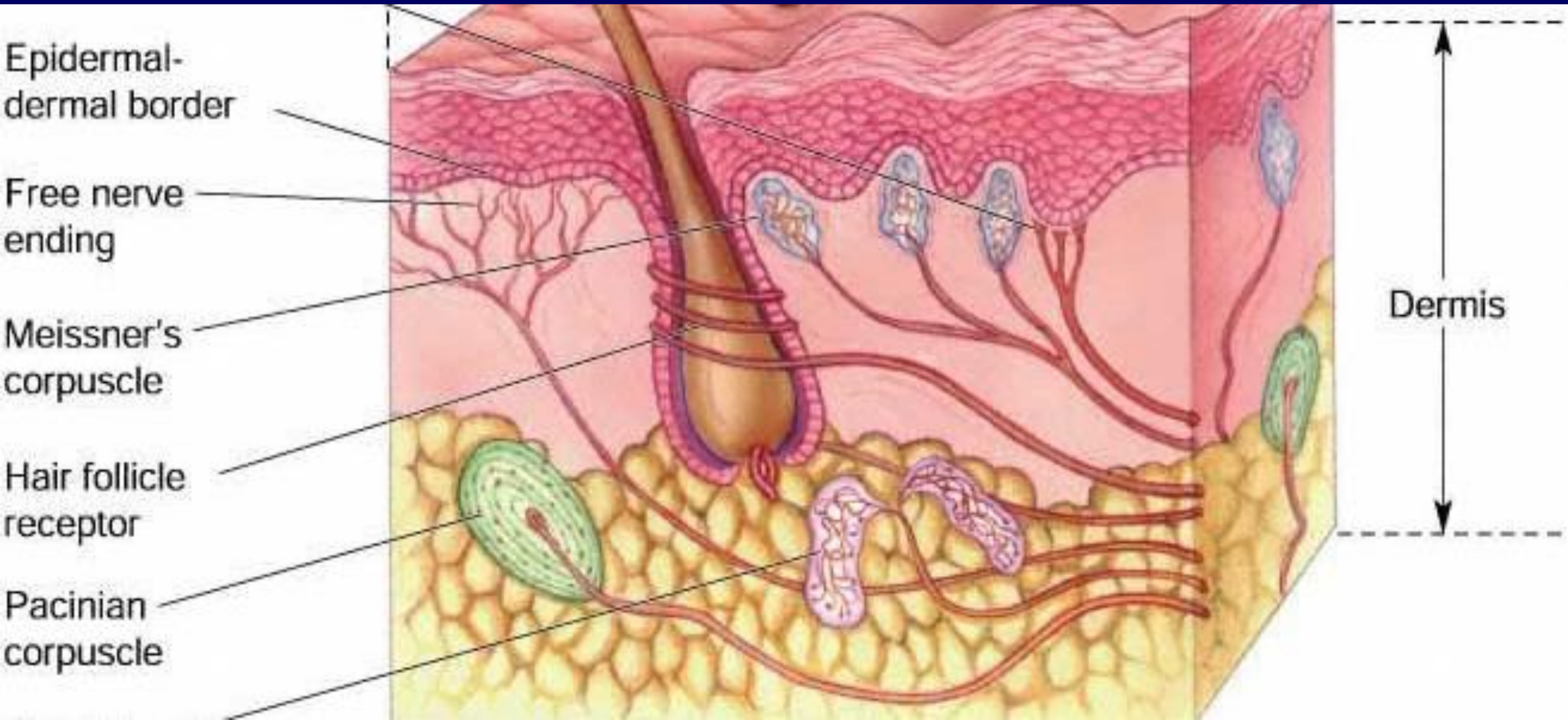
C: Meissner’s corpuscle (touch)

D: Muscle spindle (stretch)

E: Ruffini's endings respond to tension and stretch in the skin

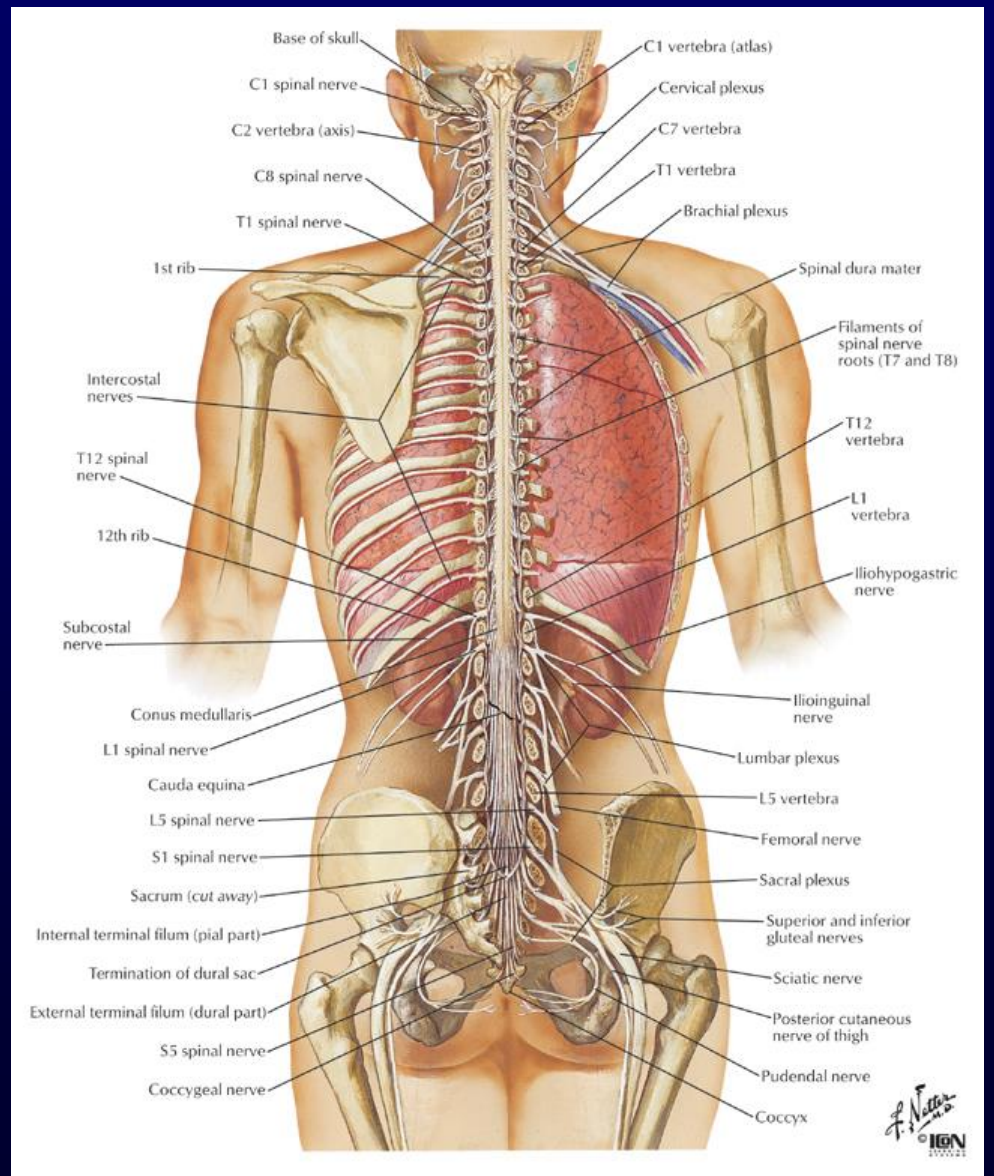


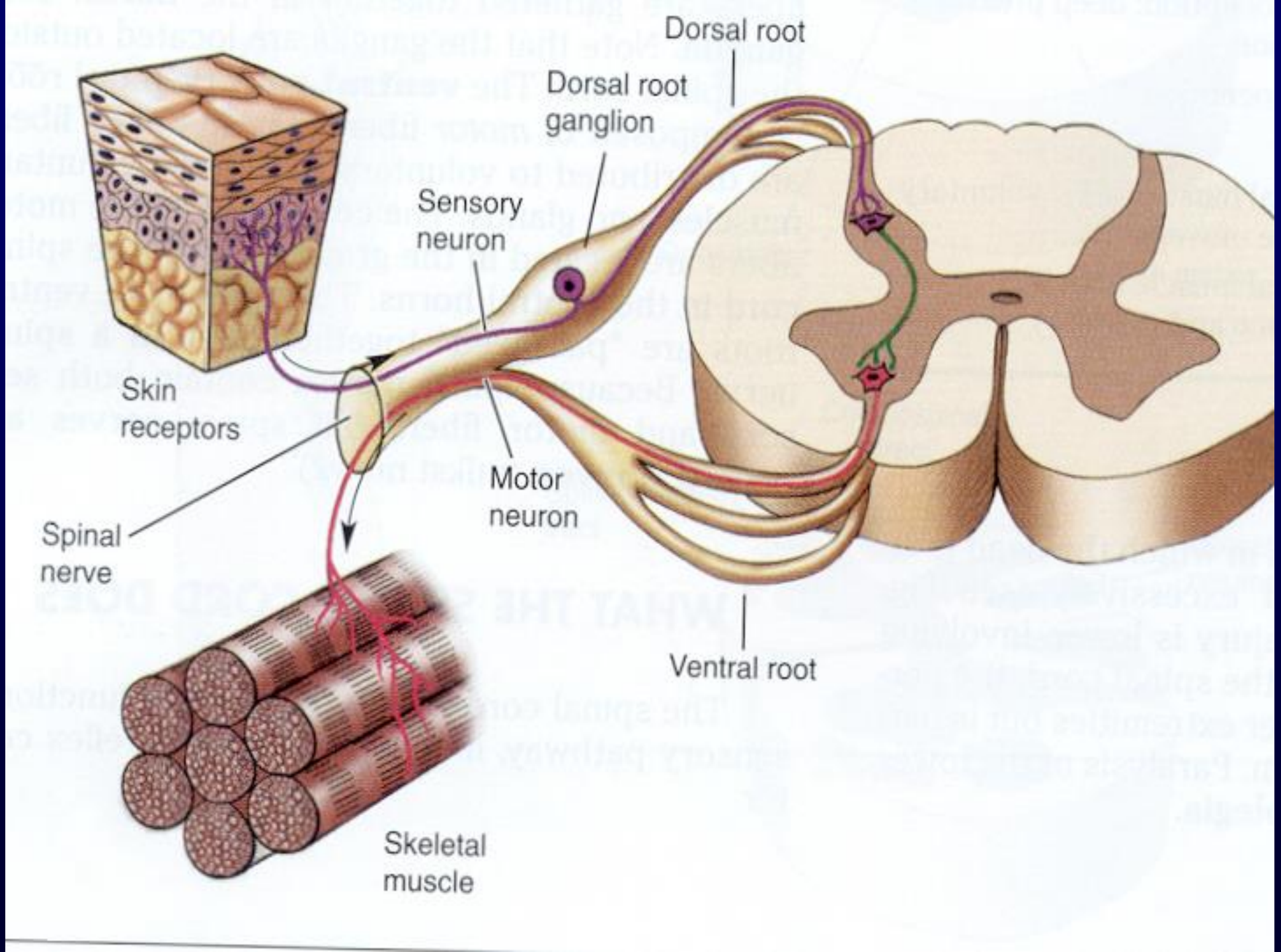
Ruffini's endings respond to tension and stretch in the skin



Spinal Cord

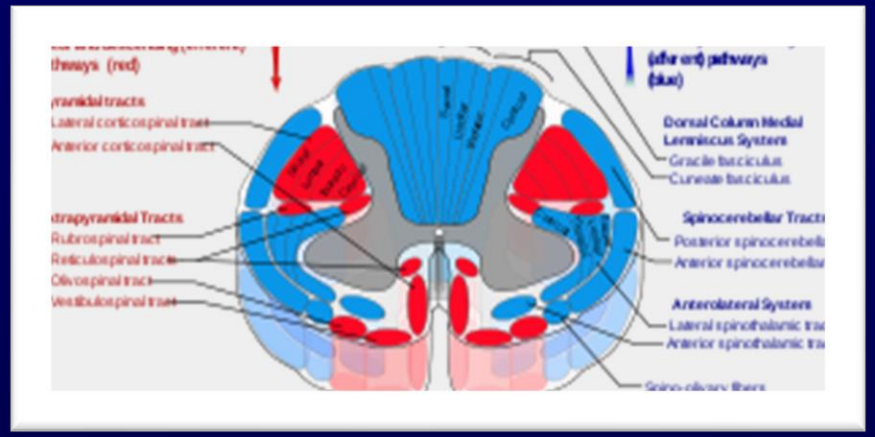
- Multiple ascending and descending tracts and interneurons (connect afferent & efferent)
- **Afferent Neurons – Enter to dorsal (back) side**
- **Efferent Neurons – Exit the ventral (front) side**

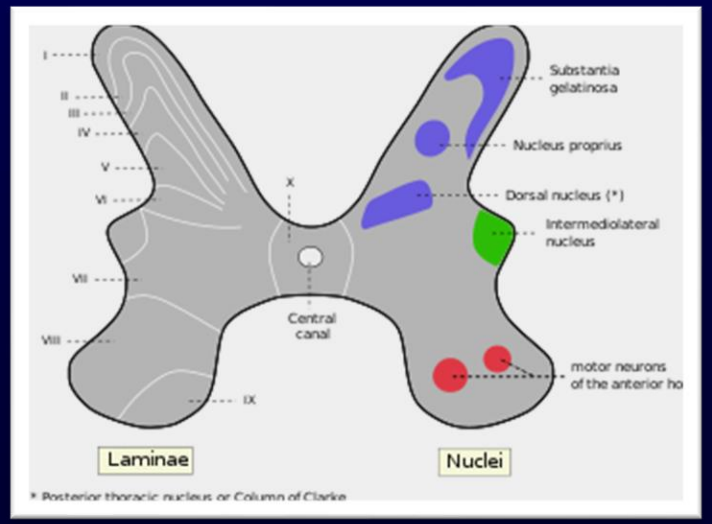



Dorsal root
Dorsal root ganglion
Sensory neuron
Skin receptors
Spinal nerve
Motor neuron
Ventral root
Skeletal muscle

Spinal Cord

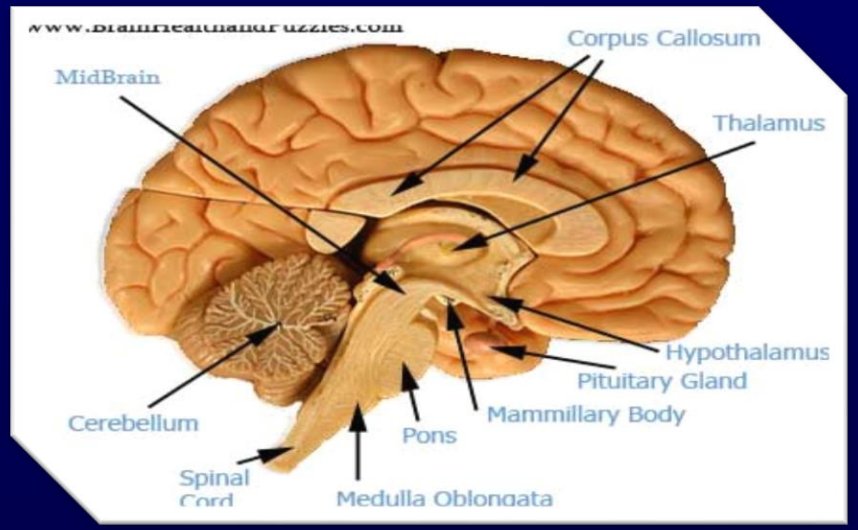
- **Spinal Layers**
  - Spinal grey matters divided into 10 layers
- **Substantia Gelatinosa**
  - Composed of a layer of cell bodies running up and down the dorsal horns of the spinal cord
  - Receive input from A and C-fibers
  - Activity in SG inhibits pain transmission



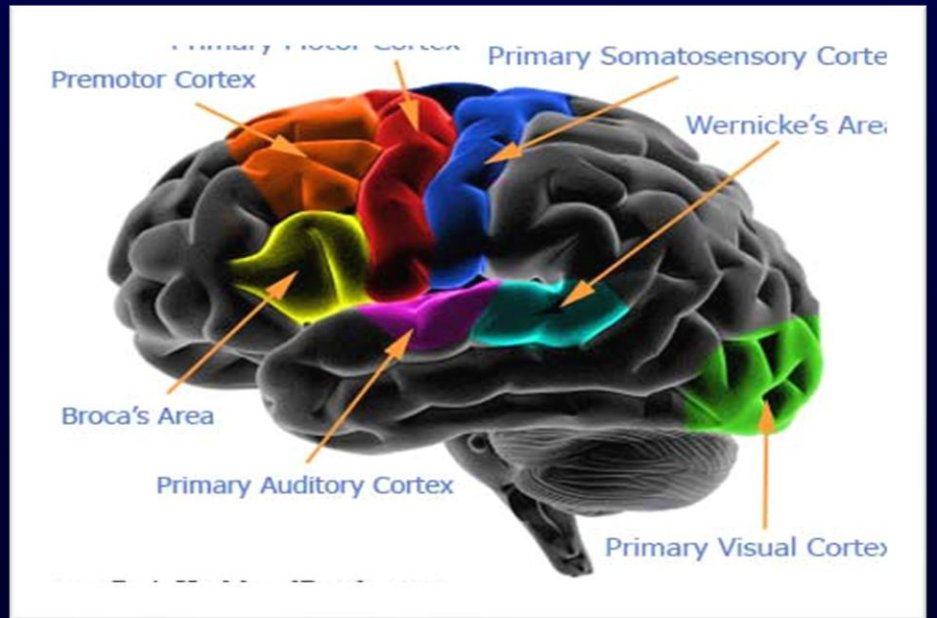


The Brain

- Thalamus



- Somatosensory Cortex

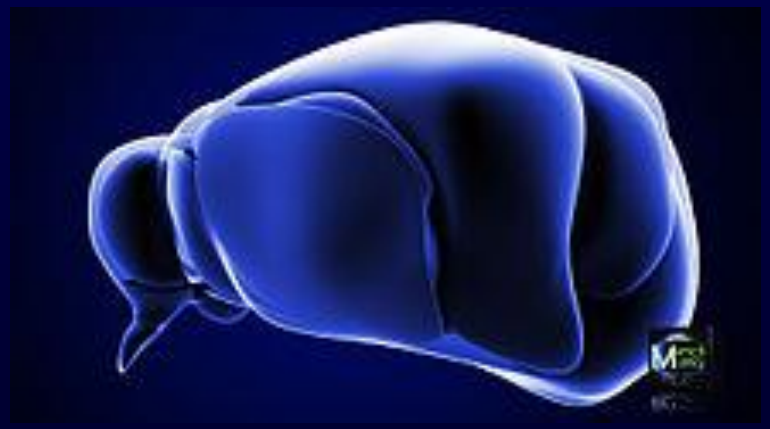


Thalamus

- The sensory switchboard of the brain



- Located in the middle of the brain

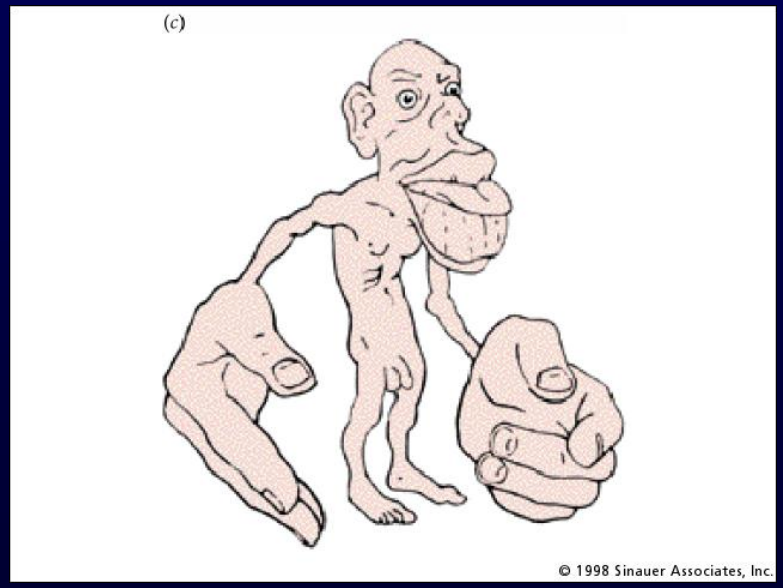


Somatosensory Cortex

- Area of cerebral cortex located in the parietal lobe right behind the frontal lobe

- Receives all info on touch and pain.

- Somatotopically organized

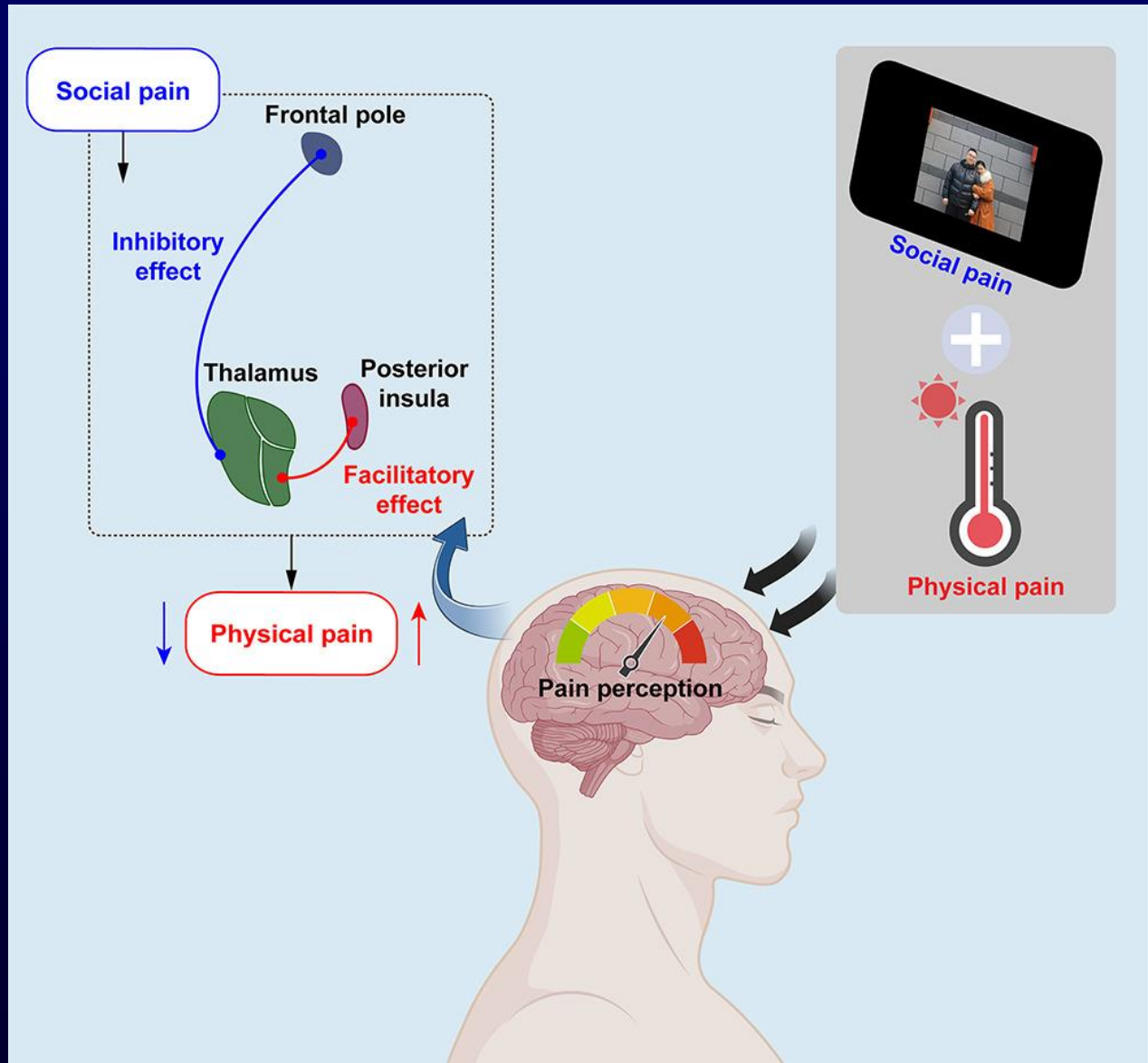


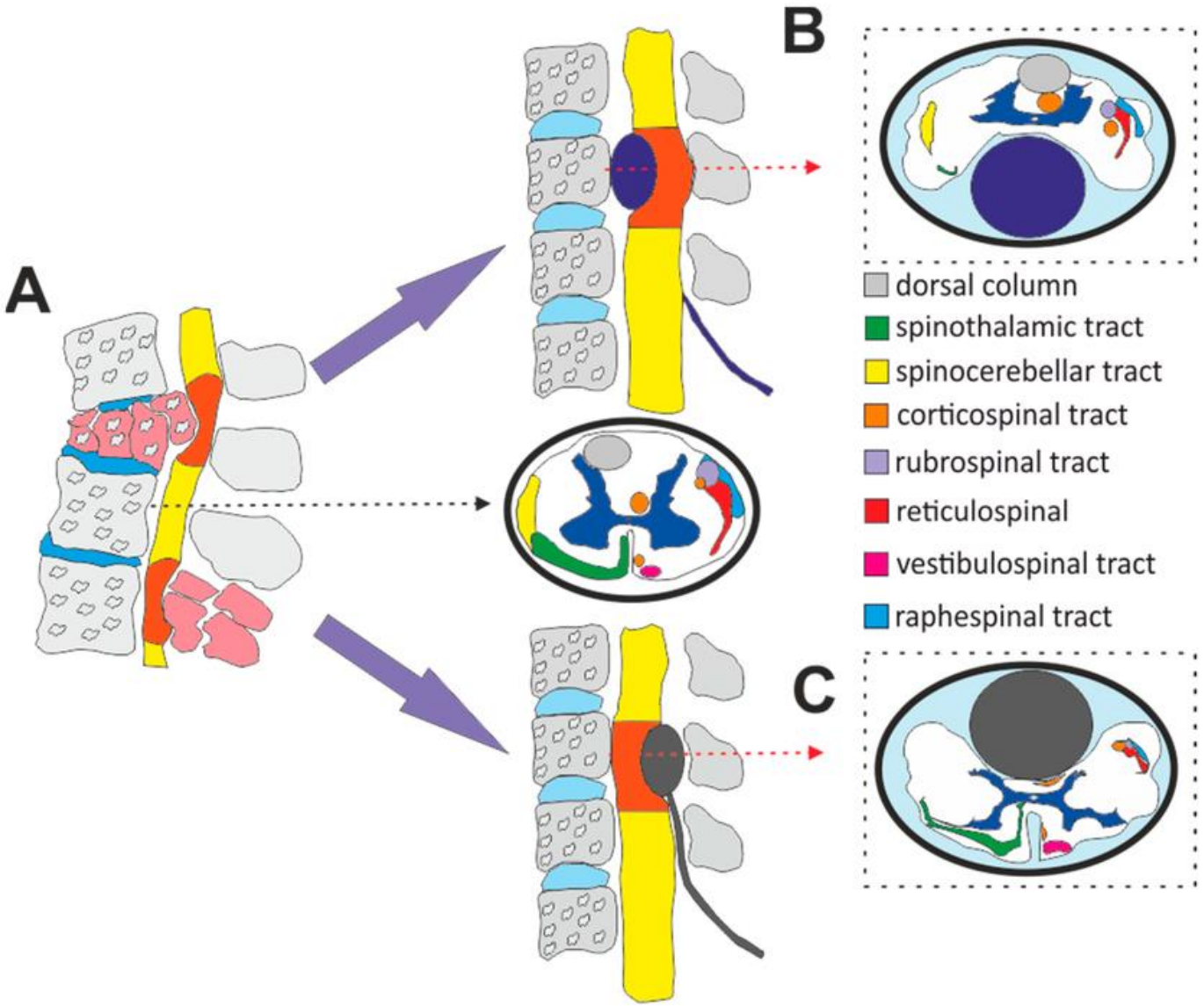

PRACTICE
MAKES
PERFECT

- Somatosensory function:
- Frontal cortex :
- Limbic system:
- Raffunii corpuscles:
- Messines's corpuscles:
- Nerve ending:
- Thalamus:
- **Substantia Gelatinosa:**

2. Sensory pathways: 3 neurons

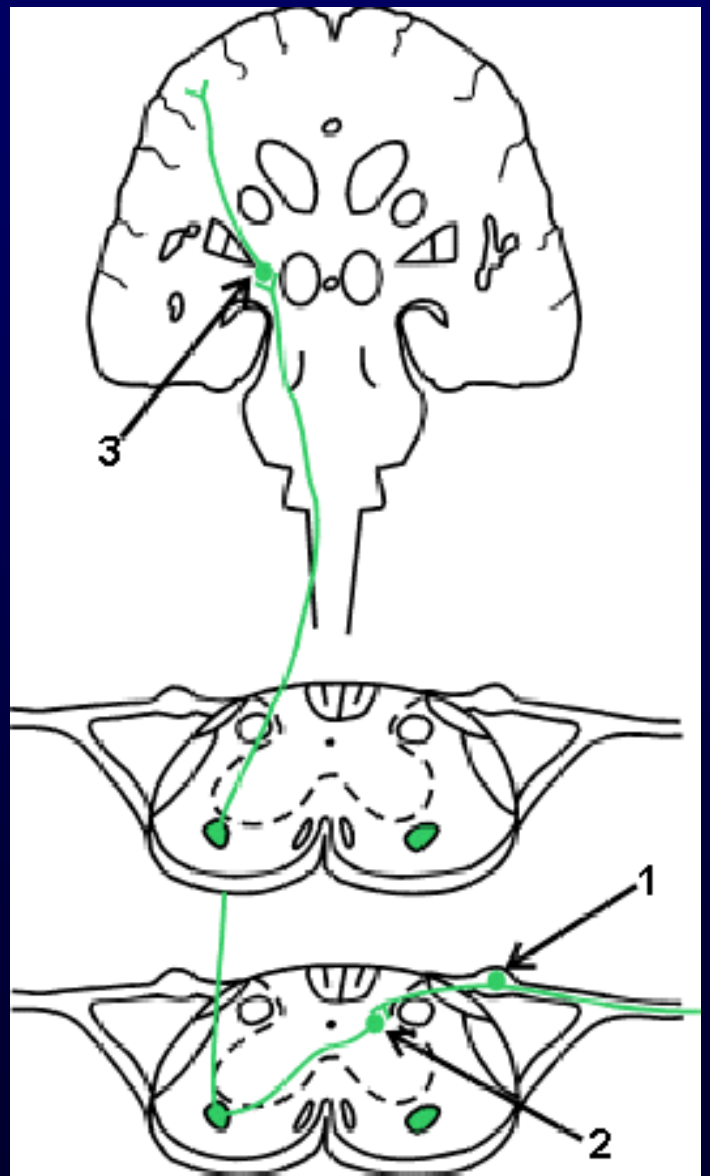
- 1st: enters spinal cord from periphery
- 2nd: crosses over (decussates), ascends in spinal cord to thalamus
- 3rd: projects to somatosensory cortex

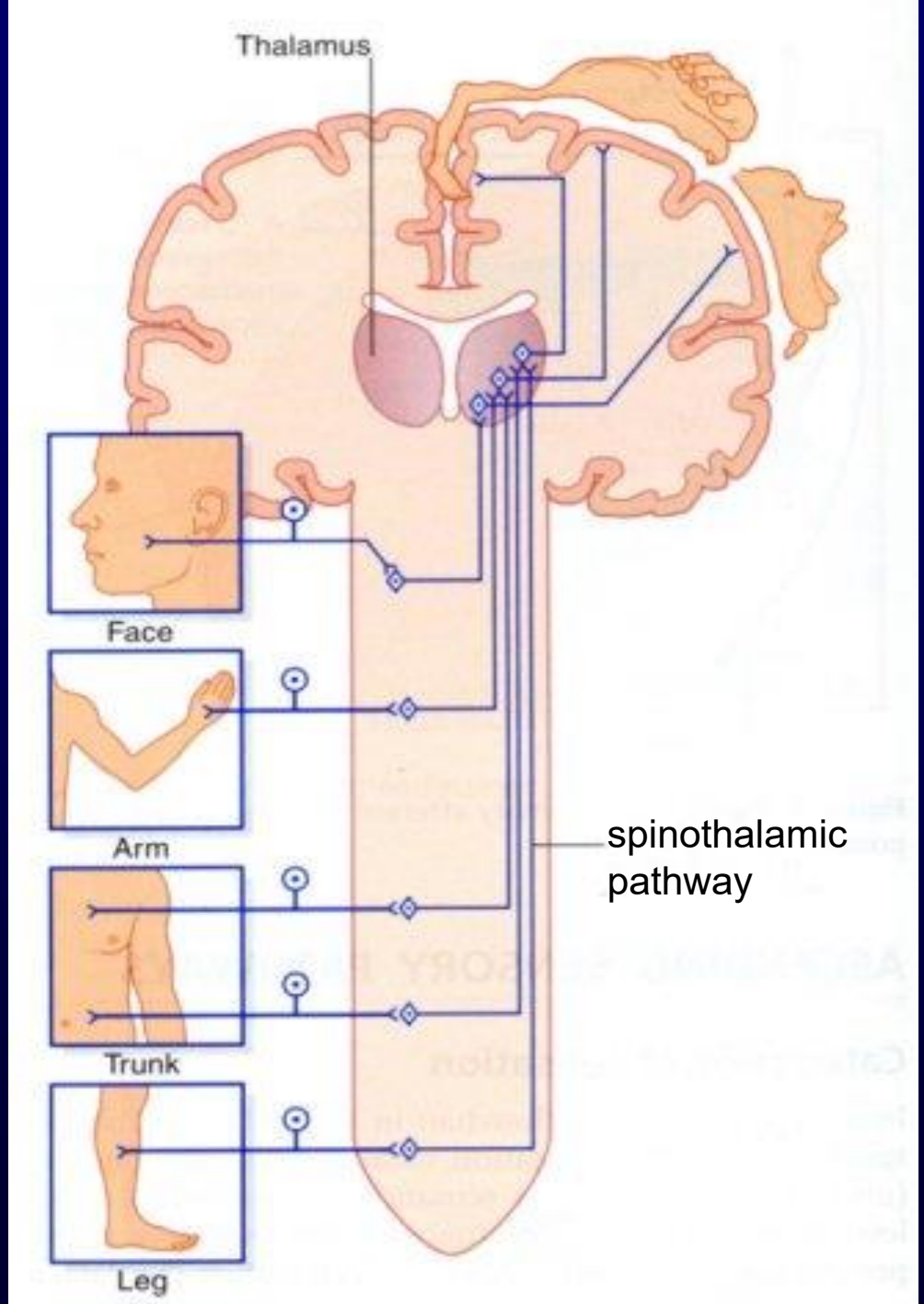



A
B
C
dorsal column
spinothalamic tract
spinocerebellar tract
corticospinal tract
rubrospinal tract
reticulospinal
vestibulospinal tract
raphespinal tract

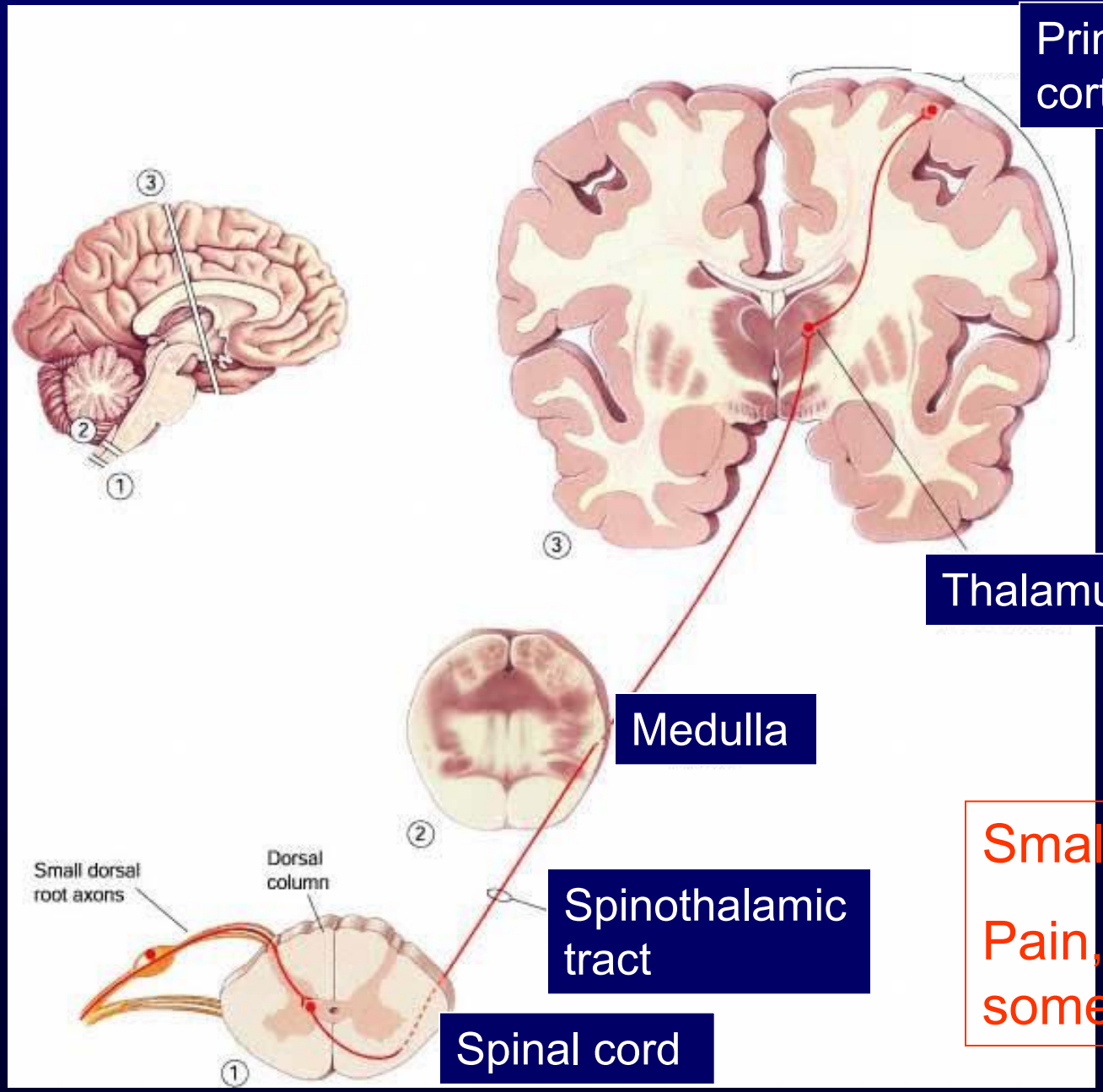
2.1 Spinothalamic pathway

- **Carries pain, temperature, touch and pressure signals**
- **1st neuron** enters spinal cord through dorsal root
- **2nd neuron** crosses over in spinal cord; ascends to thalamus
- **3rd neuron** projects from thalamus to somatosensory cortex




Thalamus
Face
Arm
Trunk
Leg
spinothalamic pathway

Spinothalamic Pathway



Primary somatosensory cortex (S1)

Thalamus

Medulla

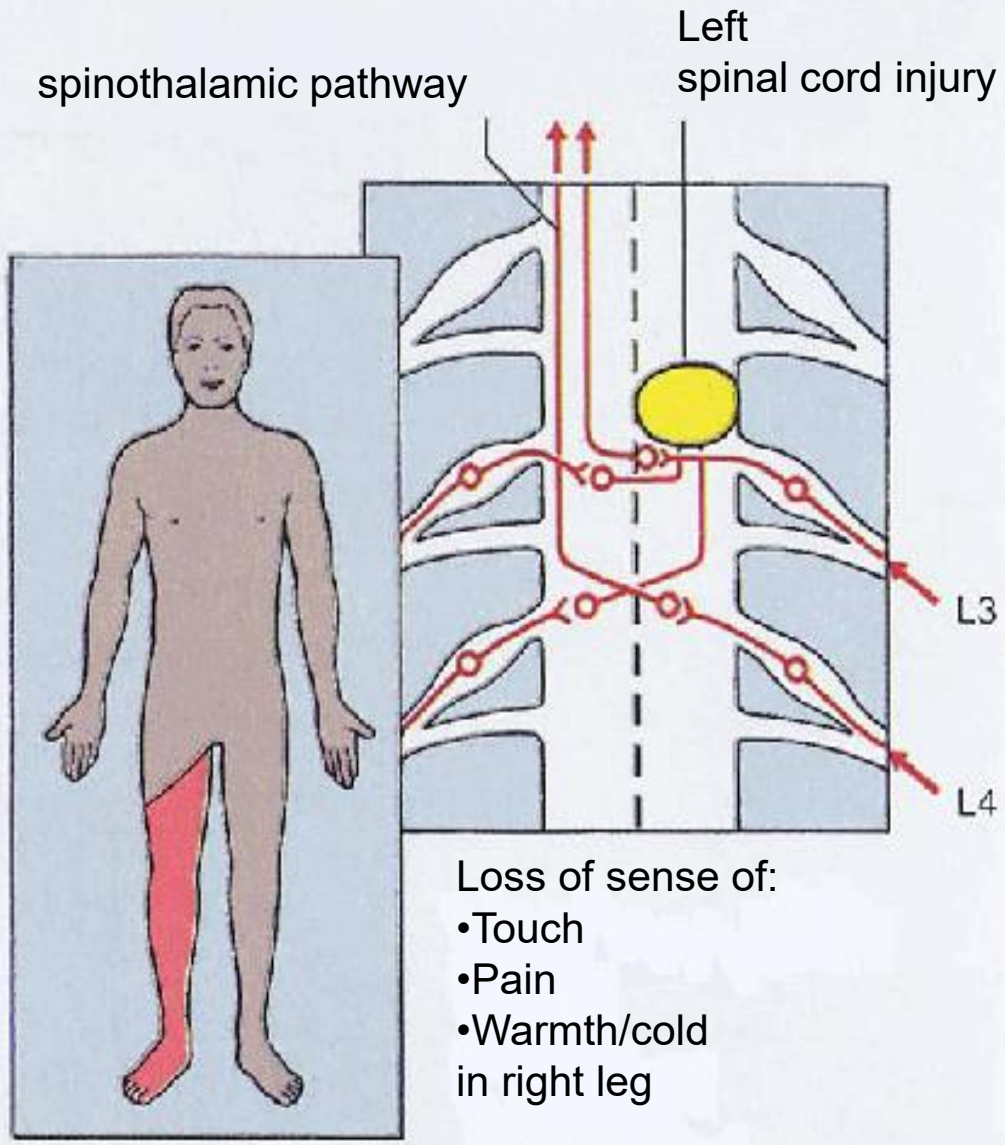
Spinothalamic tract

Spinal cord

Small sensory fibres:

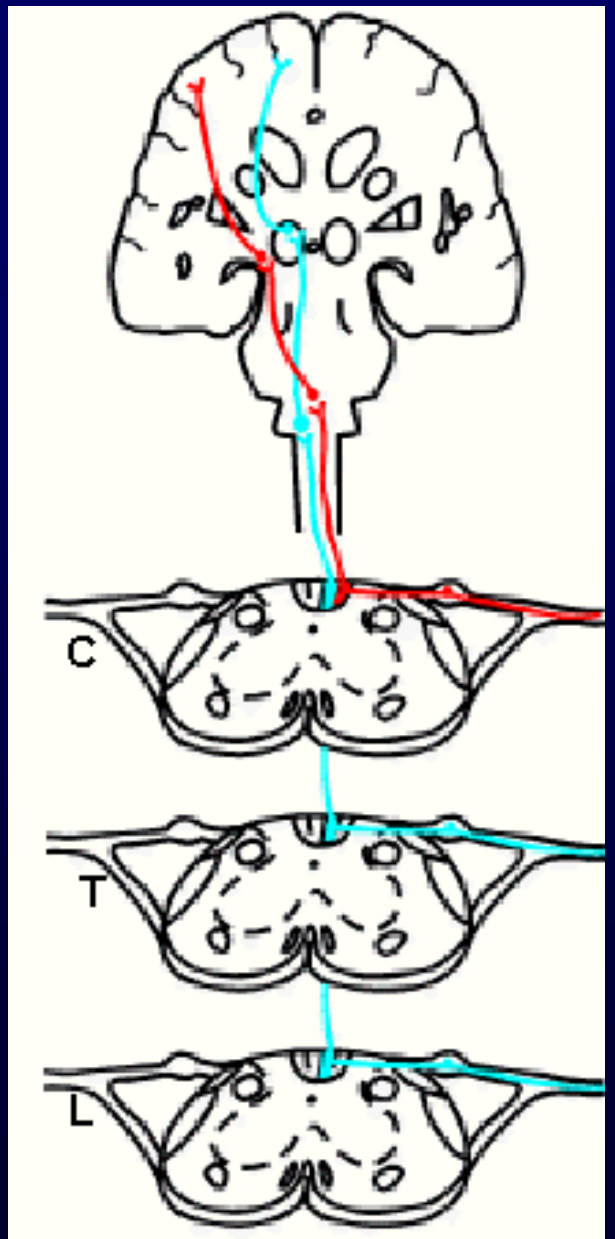
Pain, temperature, some touch

Spinothalamic damage

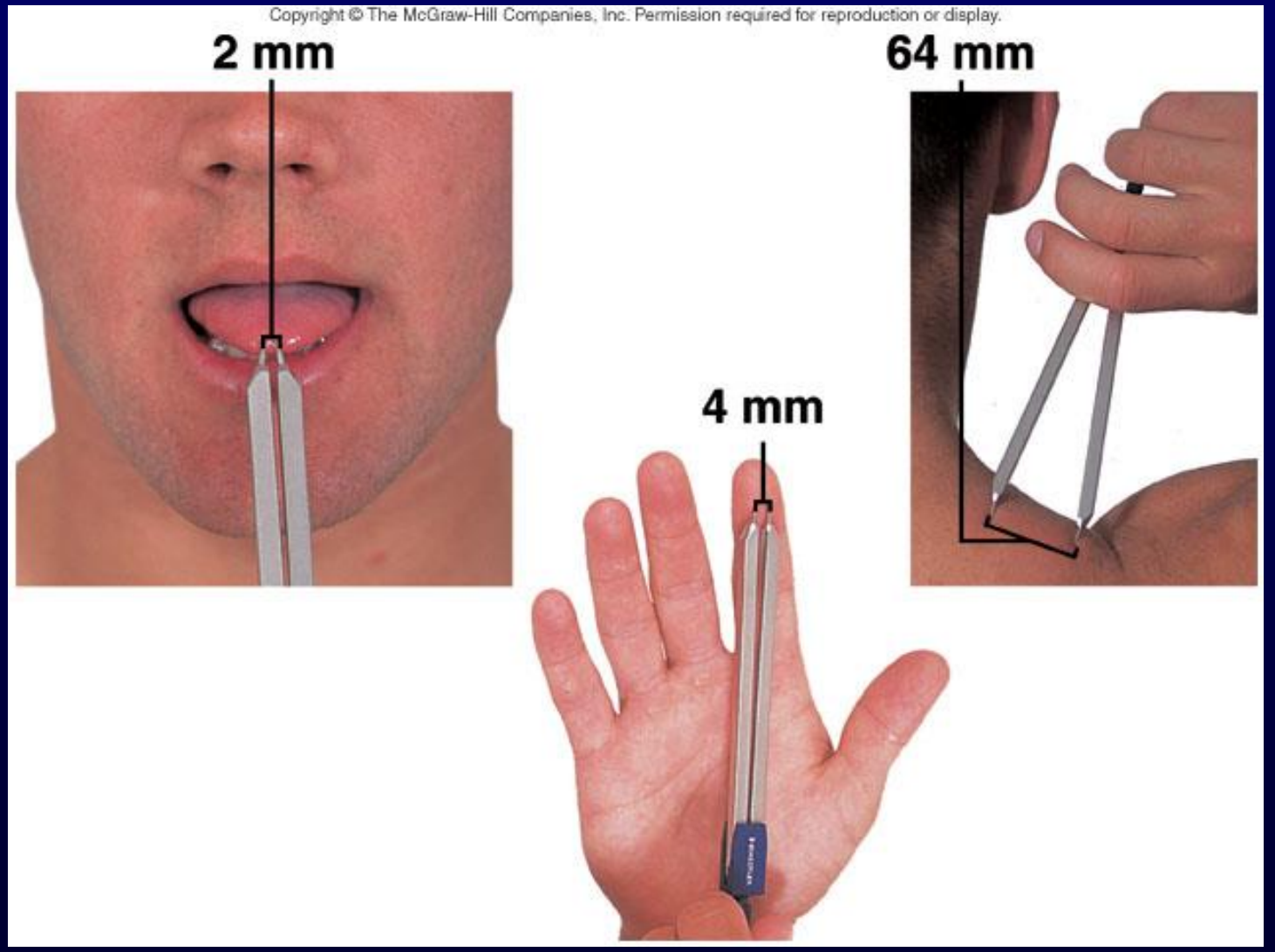


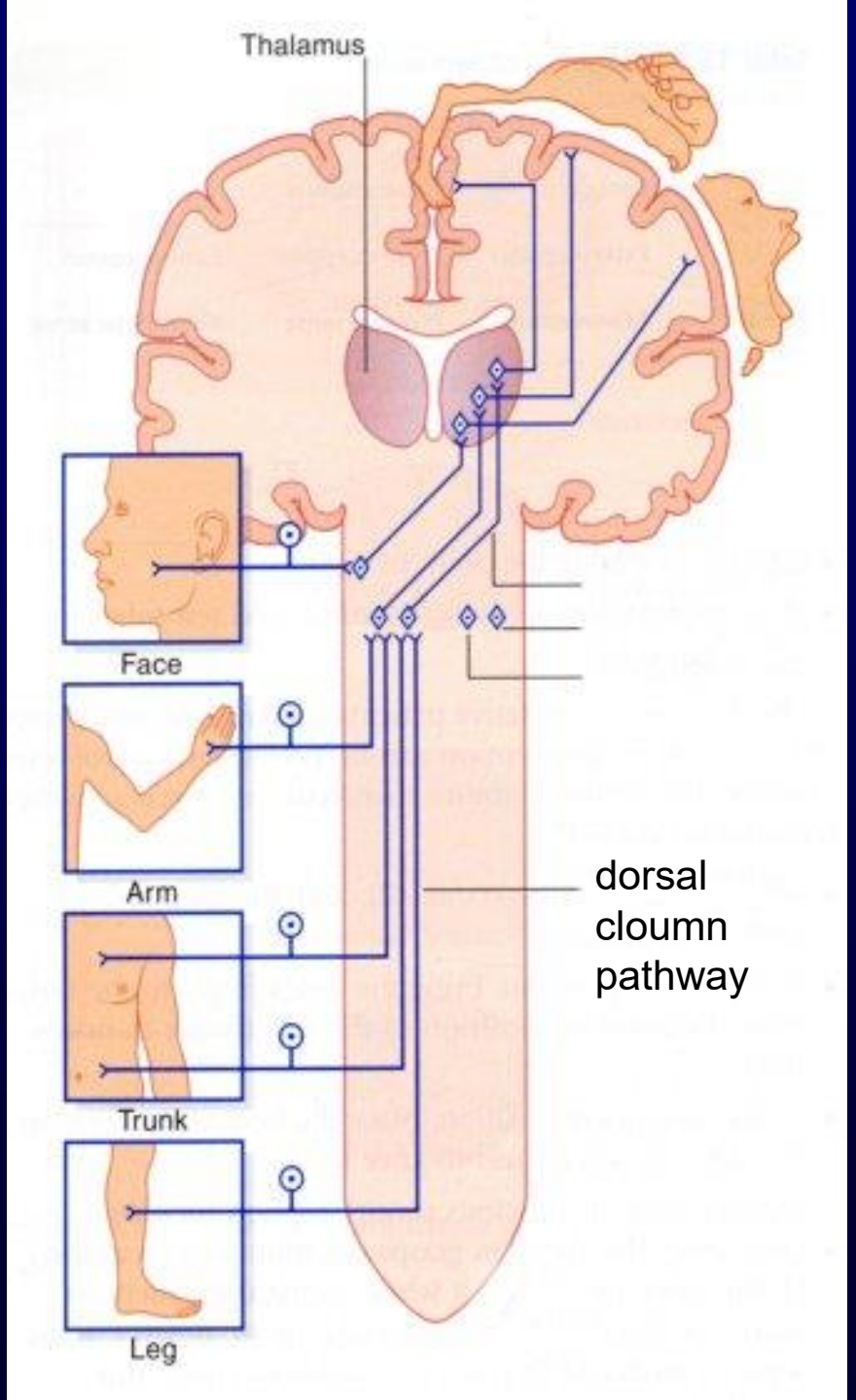
2.2 Dorsal column pathway

- **Carries fine touch, vibration and conscious proprioception signals**
- 1st neuron enters spinal cord through dorsal root; ascends to medulla (brain stem)
- 2nd neuron crosses over in medulla; ascends to thalamus
- 3rd neuron projects to somatosensory cortex

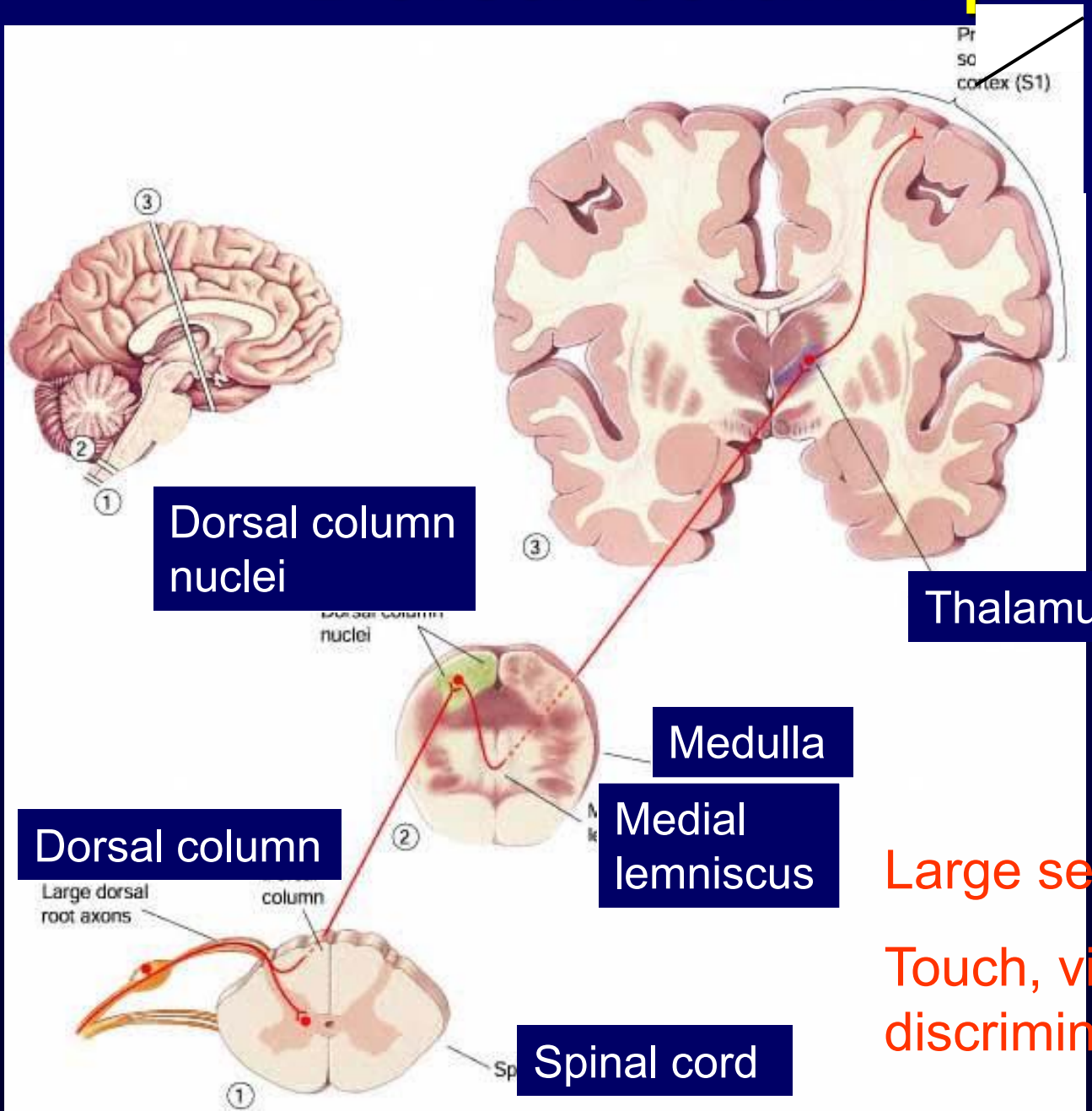


Two-Point Discrimination

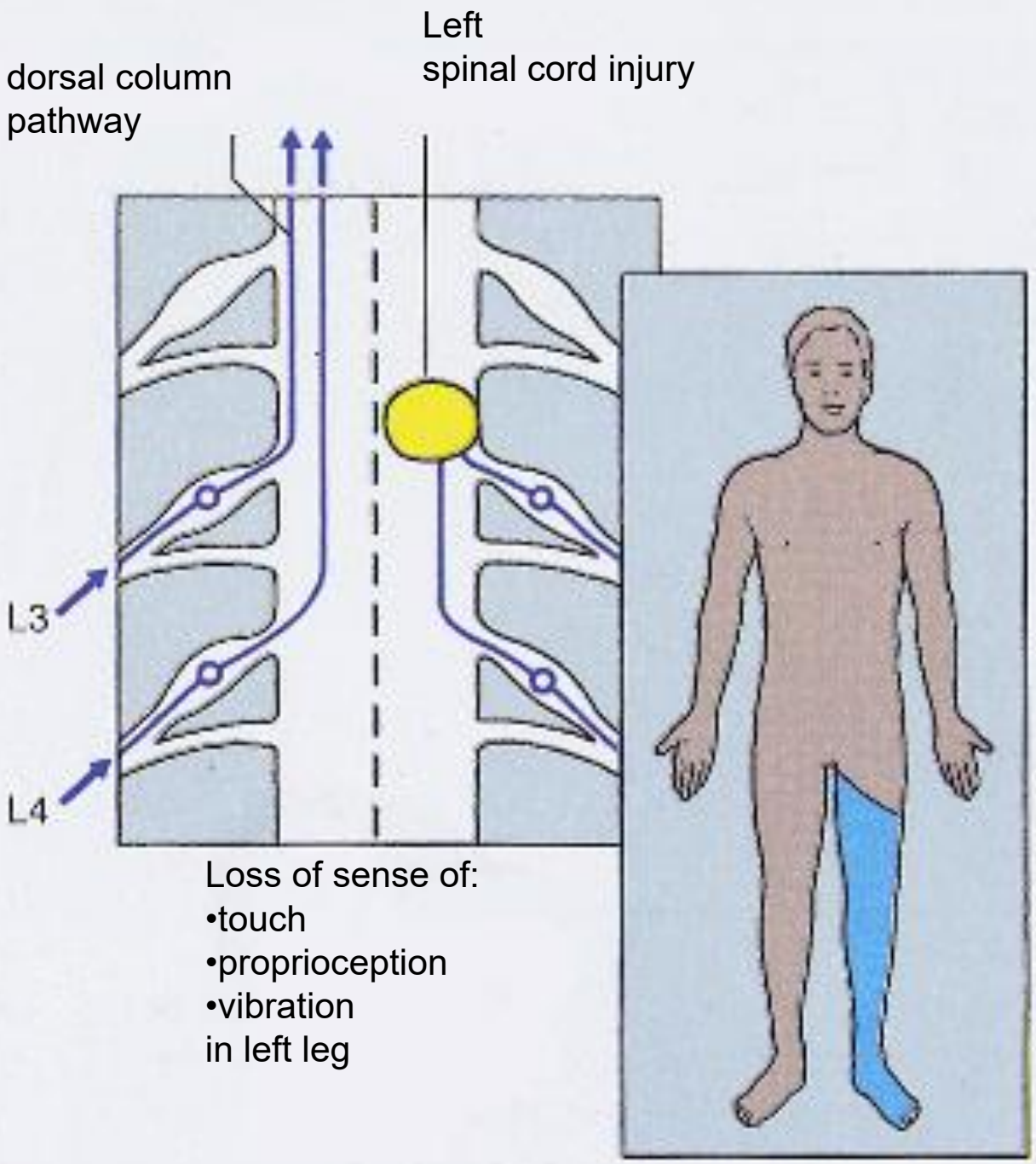



Thalamus
Face
Arm
Trunk
Leg
dorsal
cloumn
pathway

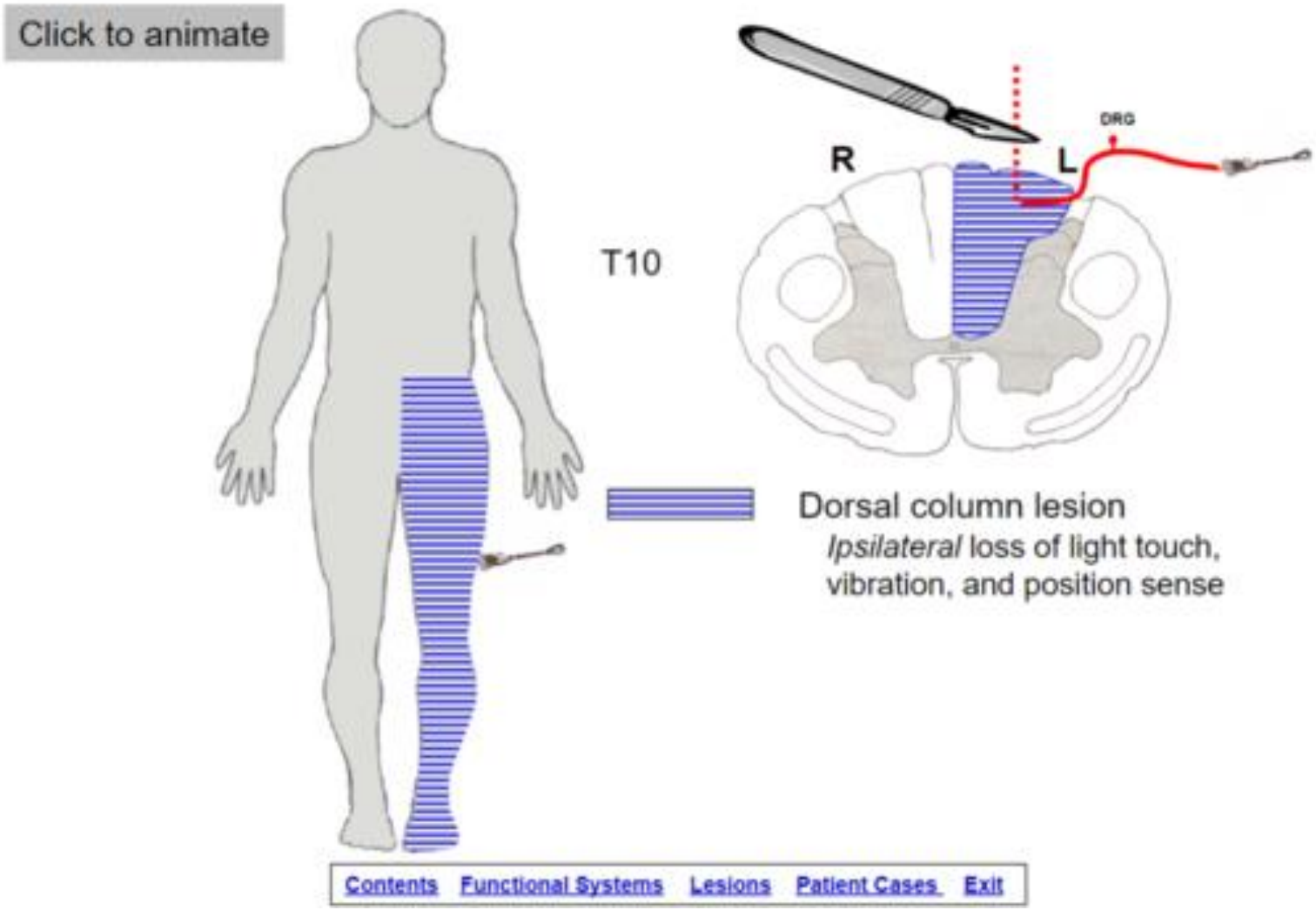
Dorsal column pathway



Dorsal column damage



Dorsal column damage

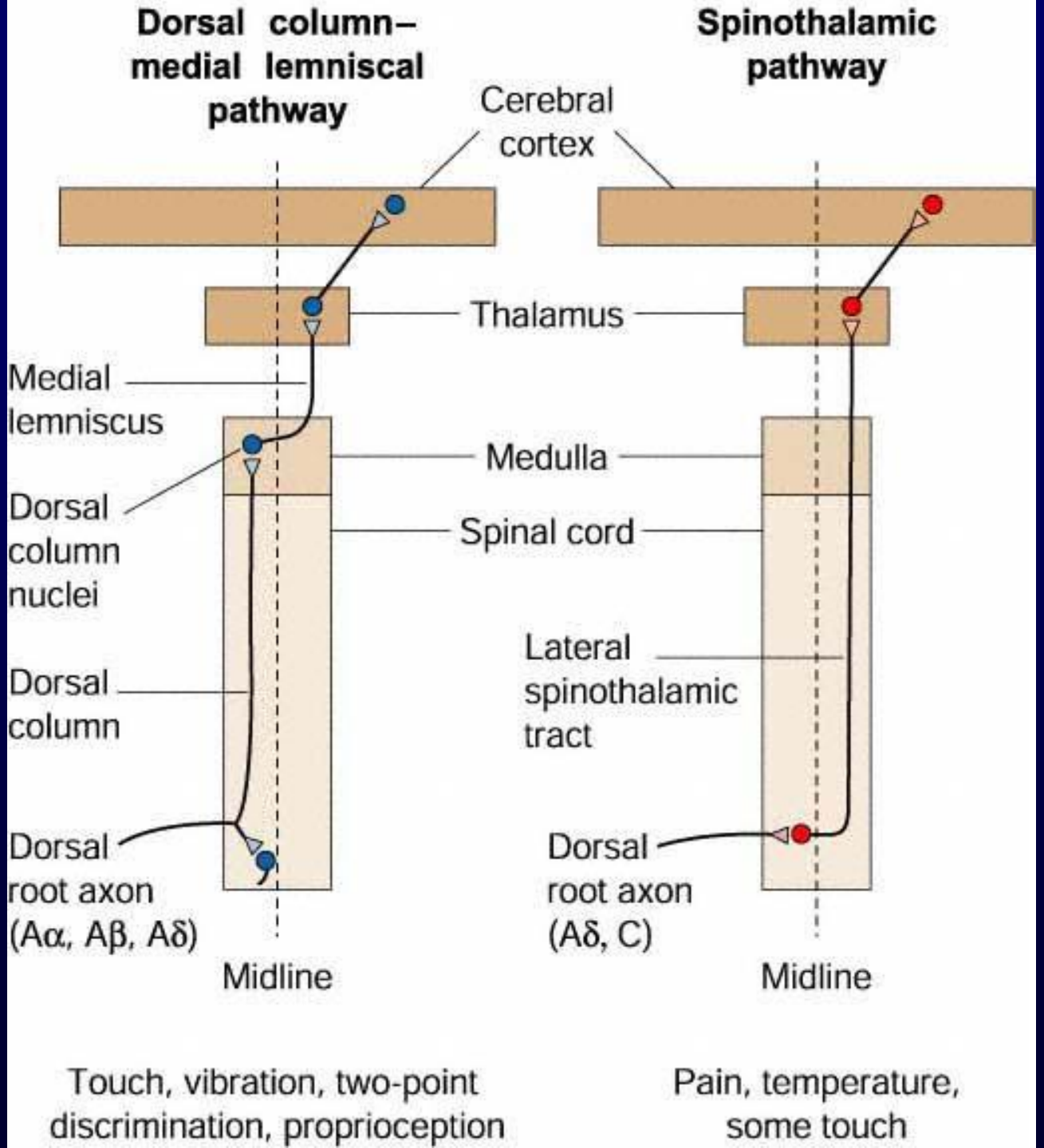


Dorsal column damage

- Sensory ataxia
- Patient staggers; cannot perceive position or movement of legs
- Visual clues help movement

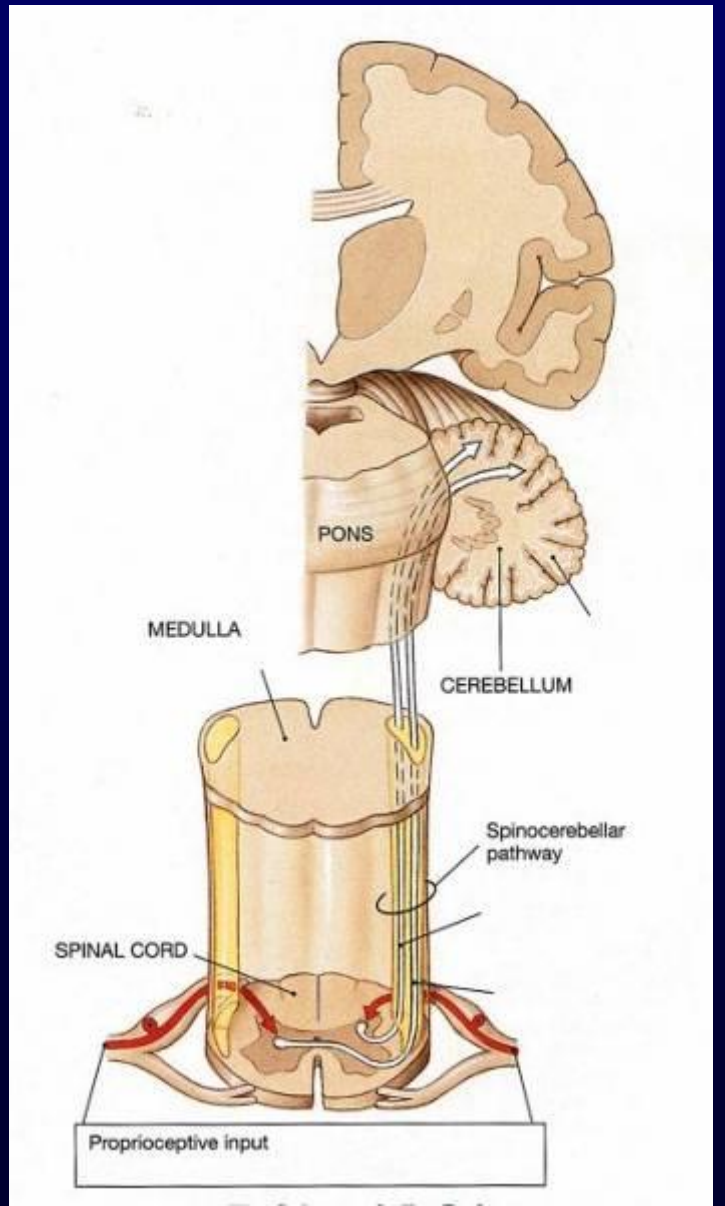


Central Pathways



3.3 Spinocerebellar pathway

- Carries unconscious proprioception signals
- Receptors in muscles & joints
- 1st neuron: enters spinal cord through dorsal root
- 2nd neuron: ascends to cerebellum
- No 3rd neuron to cortex, hence unconscious

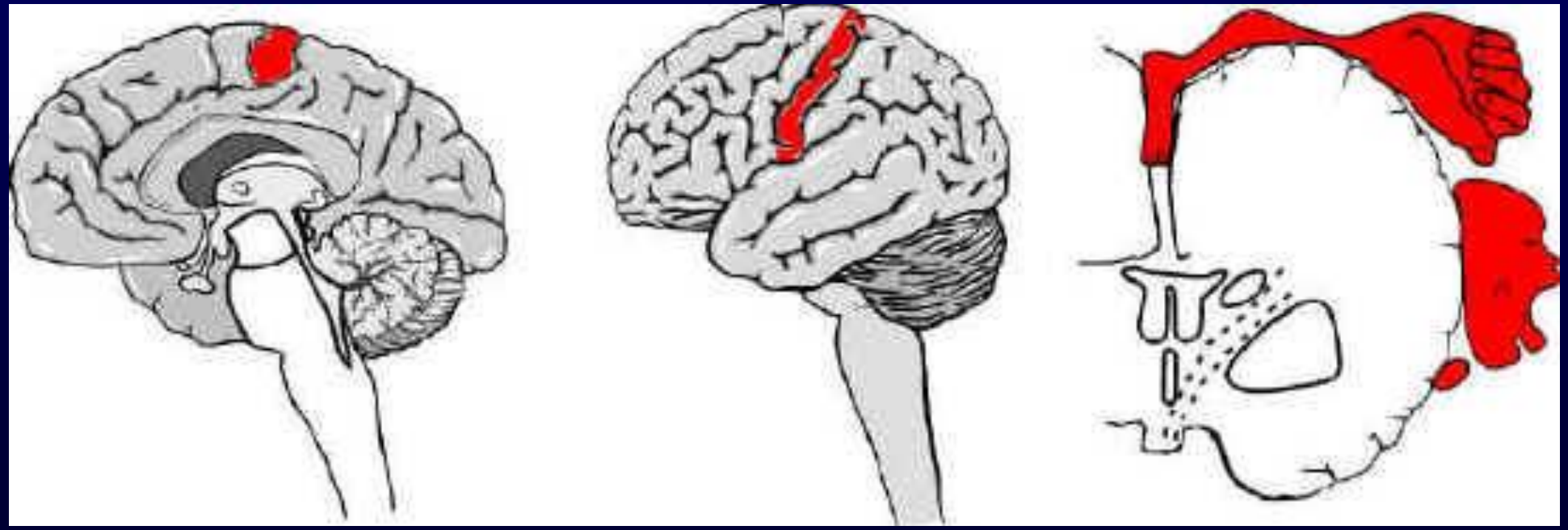


Spinocerebellar tract damage

- Cerebellar ataxia
- Clumsy movements
- Incoordination of the limbs (intention tremor)
- Wide-based, reeling gait (ataxia)
- Alcoholic intoxication produces similar effects!

4. Somatosensory cortex

Located in the postcentral gyrus of the human cerebral cortex.

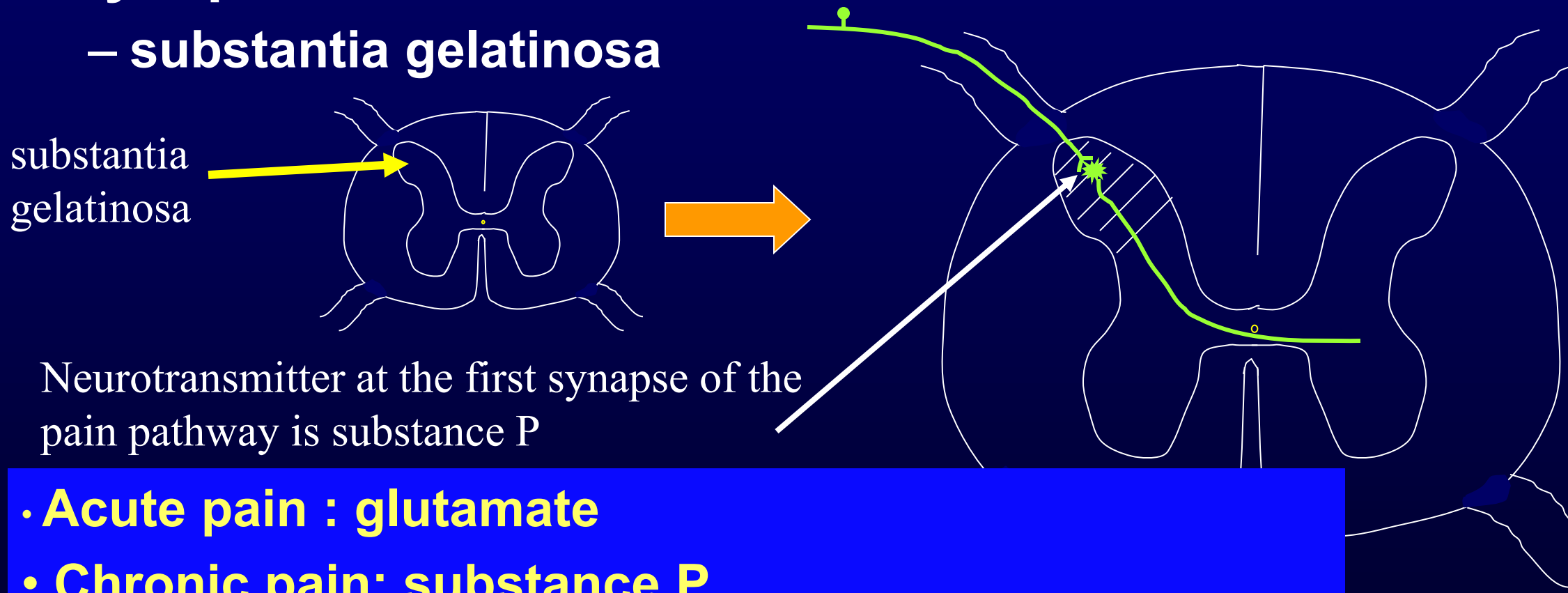


afferent fibres

- two types
  - Aδ (thin myelinated)
  - C (unmyelinated)

central connections

- afferent fibre enters the spinal cord
- synapses in laminae ii,iii
  - substantia gelatinosa

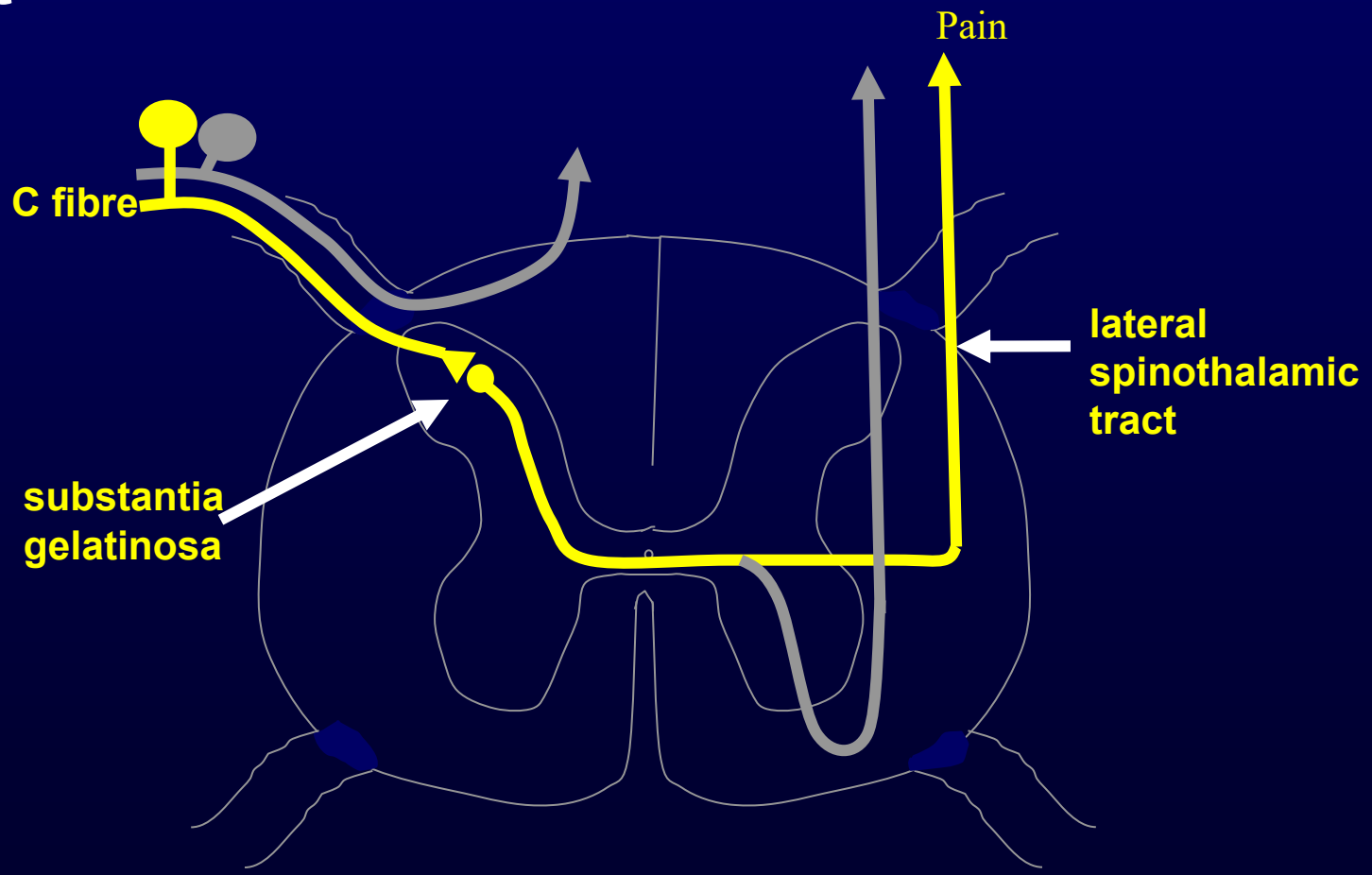


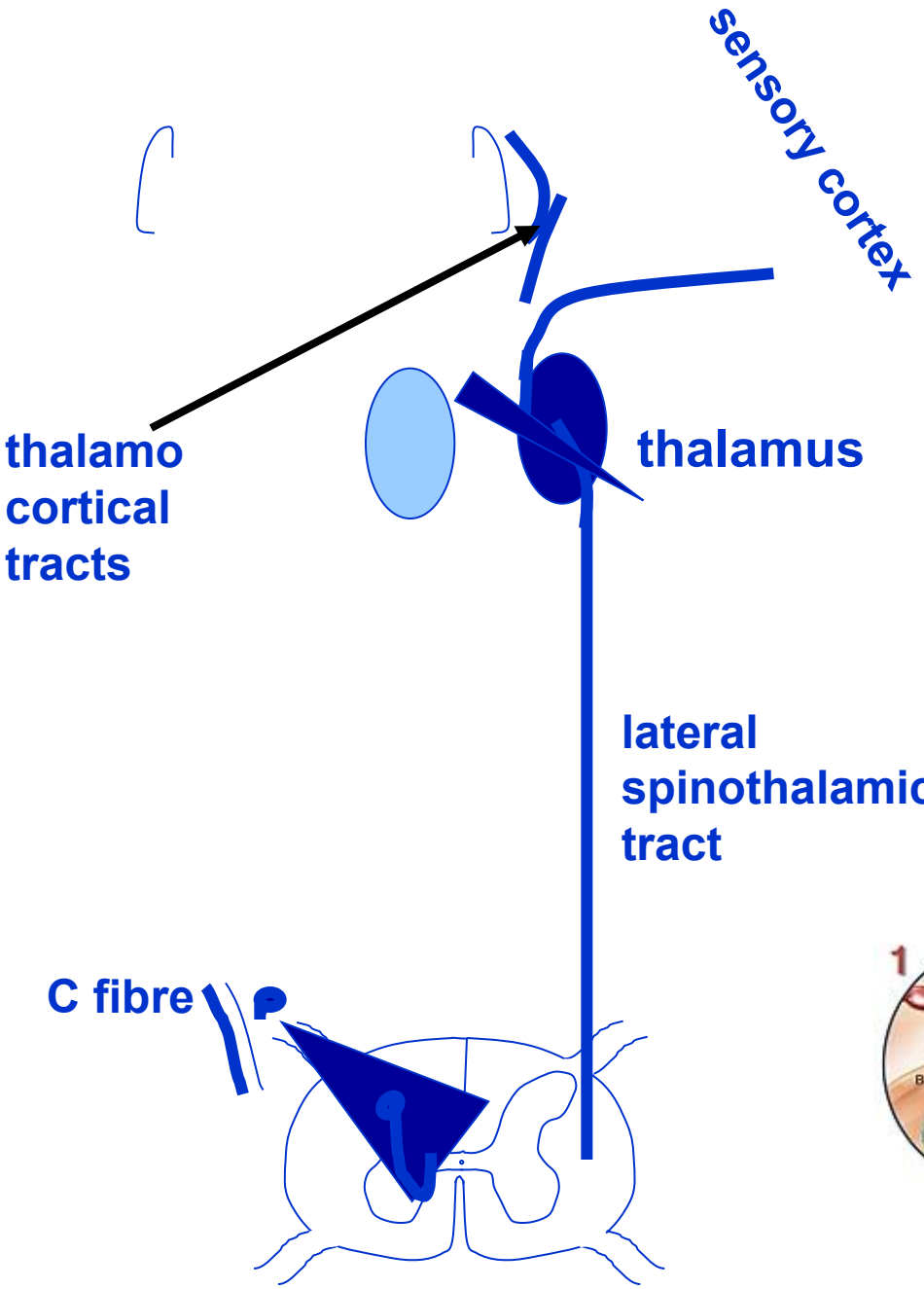
Neurotransmitter at the first synapse of the pain pathway is substance P

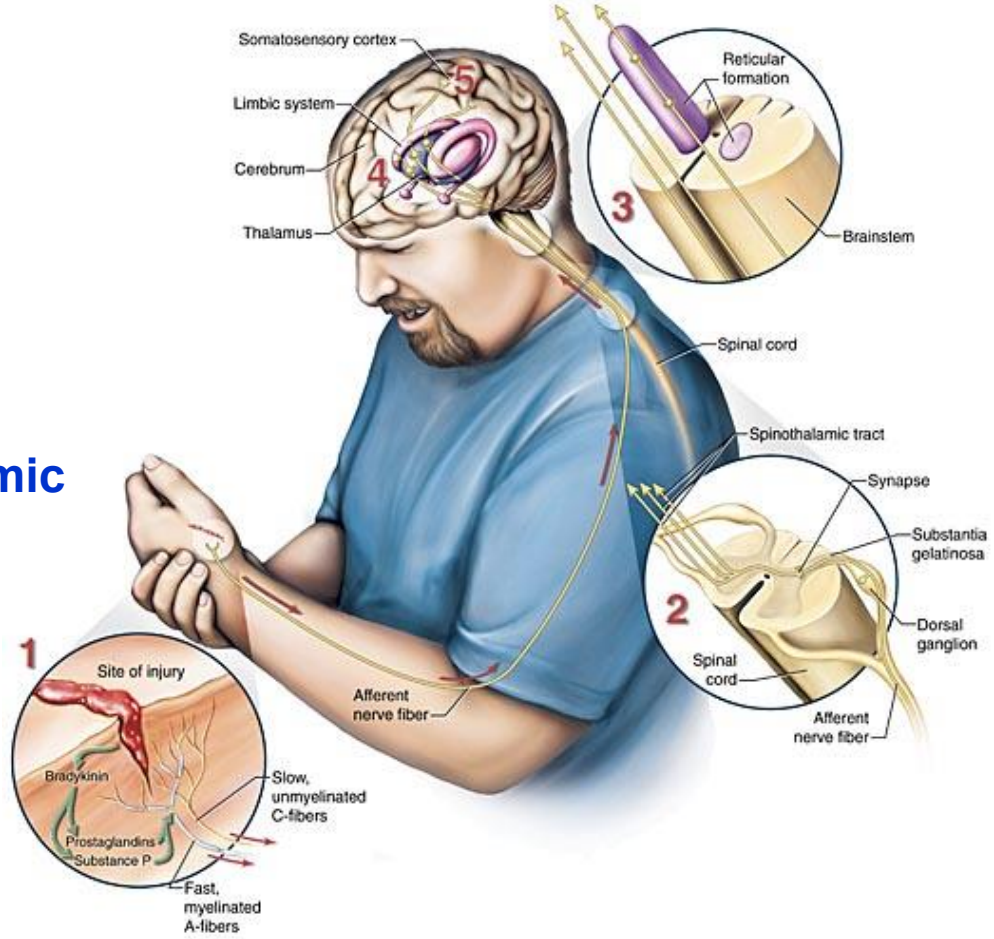
- Acute pain : glutamate
- Chronic pain: substance P
- Pain inhibitory neurotransmitters: enkephalin, GABA

ascending pathway

- crosses the midline
- ascends up as the lateral spinothalamic tract




sensory cortex
thalamo
cortical
tracts
thalamus
lateral
spinothalamic
tract
C fibre


Somatosensory cortex
Limbic system
Cerebrum
Thalamus
Reticular formation
Brainstem
Spinal cord
Spinothalamic tract
Synapse
Substantia gelatinosa
Dorsal ganglion
Spinal cord
Afferent nerve fiber
Afferent nerve fiber
Site of injury
Bradykinin
Prostaglandins
Substance P
Slow, unmyelinated C-fibers
Fast, myelinated A-fibers

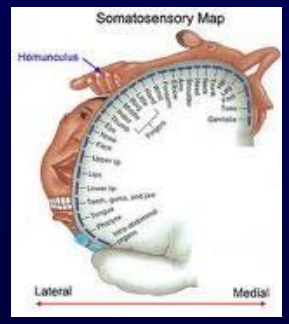
physiology of pain

Pain perception

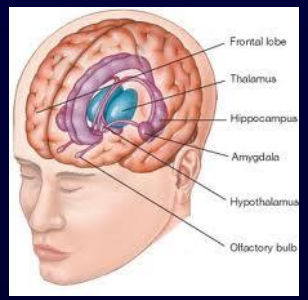
- This occurs at different levels
  - **thalamus is an important centre of pain perception**
    - **lesions of thalamus produces severe type of pain known as 'thalamic pain'**



  - **Sensory cortex is necessary for the localisation of pain**



  - **Other areas are also important**
    - **reticular formation, limbic areas, hypothalamus and other subcortical areas**



Pathophysiology of pain

- **Pain sensations could arise due to**
  - Inflammation of the nerves (neuritis)
  - Injury to the nerves and nerve endings with scar formation (disk prolapse)
  - Injury to the structures in the spinal cord, thalamus or cortical areas that process pain information (spinal trauma)
  - Abnormal activity in the nerve circuits that is perceived as pain (phantom limb pain)
  - Nerve invasion, for example by cancer (brachial plexopathy)

opioid peptides

- short peptides originally known to be secreted in CNS and later found to be present in GIT etc

opioid peptides

- **β endorphin**
  - Earliest to discover, present in pituitary
- **encephalins** - met & leu
  - widely distributed
- **dynorphin**
- **Endomorphine 1 & 2**

Receptors: kappa, delta, recently discovered ORL1 receptor

- descending tracts involving opioid peptides as neurotransmitter were discovered

- these were known to modify (inhibit) pain impulse transmission at the first synapse at the substantia gelatinosa

– it involves encephalin secreting neurons in the **reticular formation**
– starting from the PAG **(periaqueductal grey area)** of the midbrain
– ending in the NRM **(nucleus raphe magnus)** of the medulla
– from their ending in **the substantia gelatinosa** of the dorsal horn

periaqueductal grey nucleus

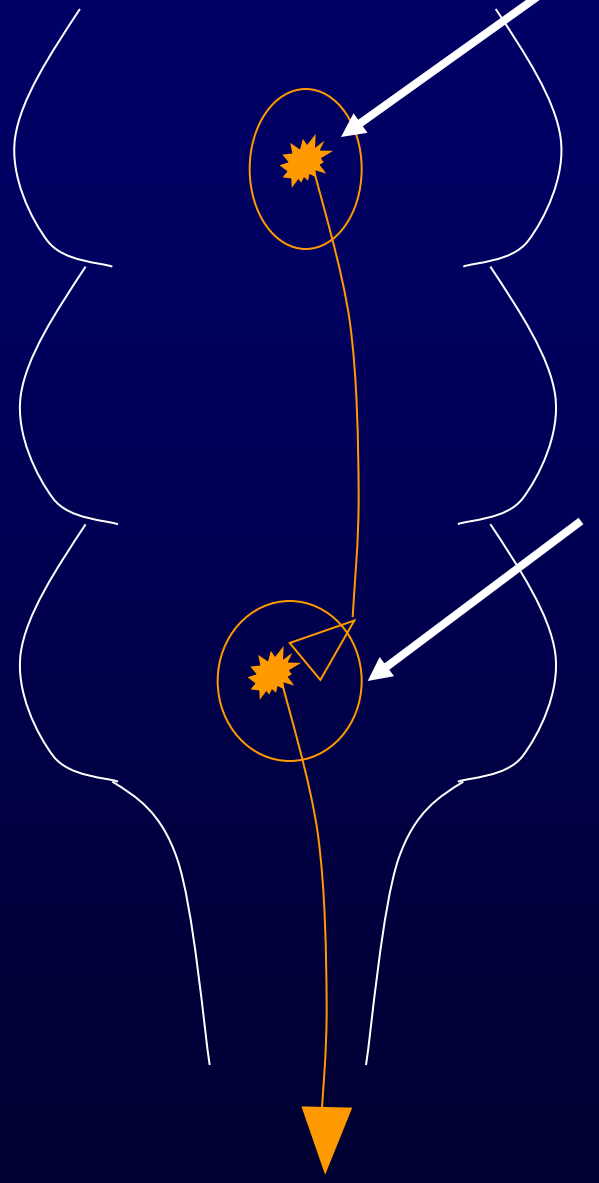
midbrain

pons

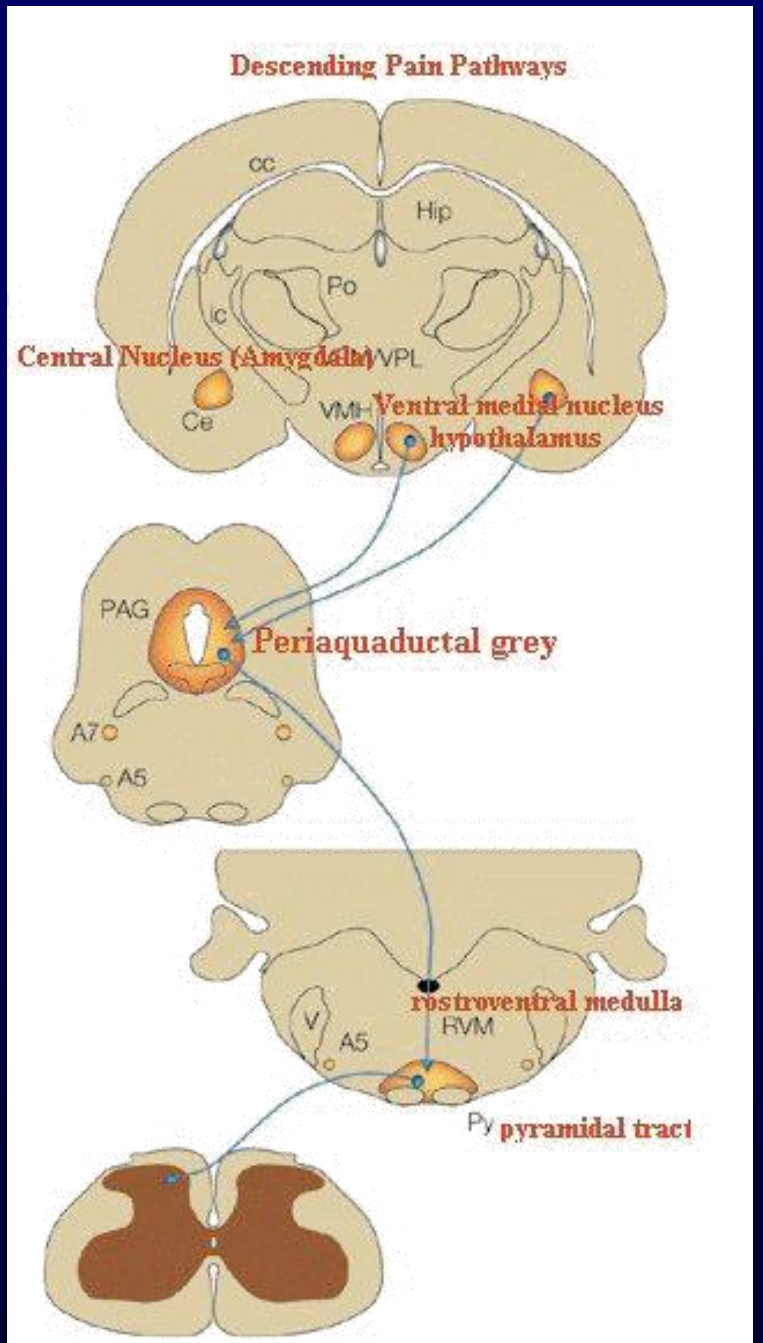
nucleus raphe magnus

medulla

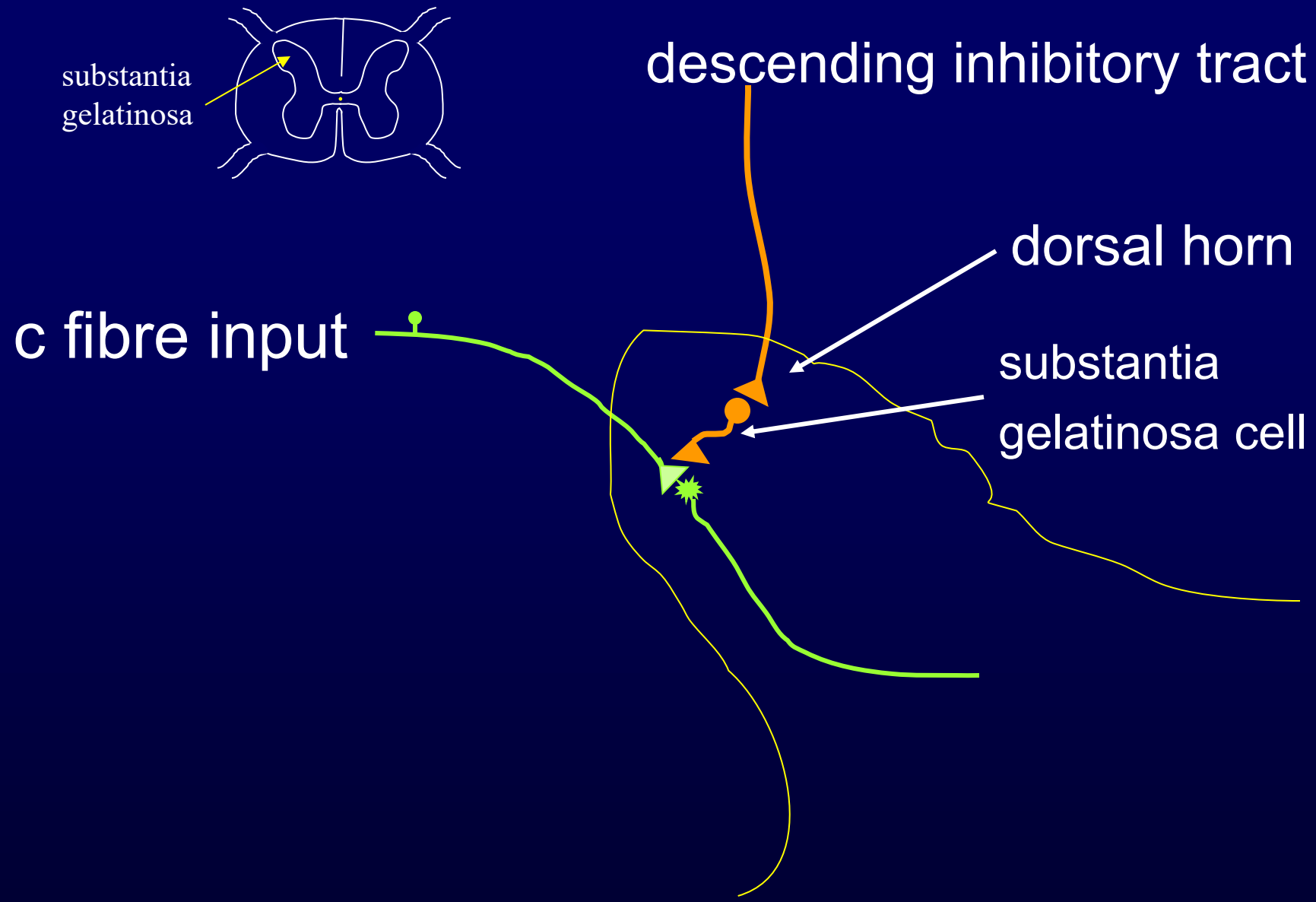
spinal cord



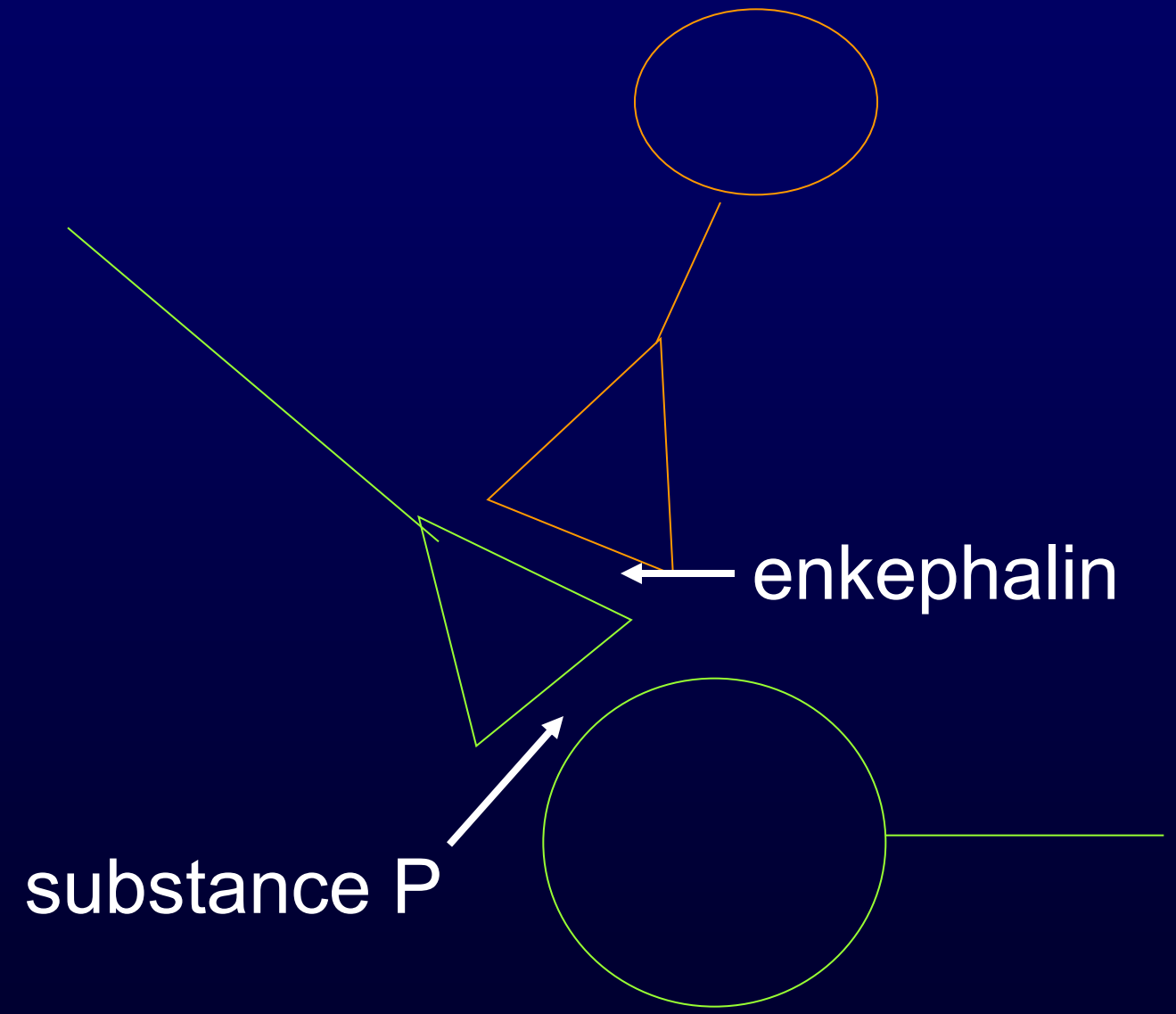
substantia gelatinosa



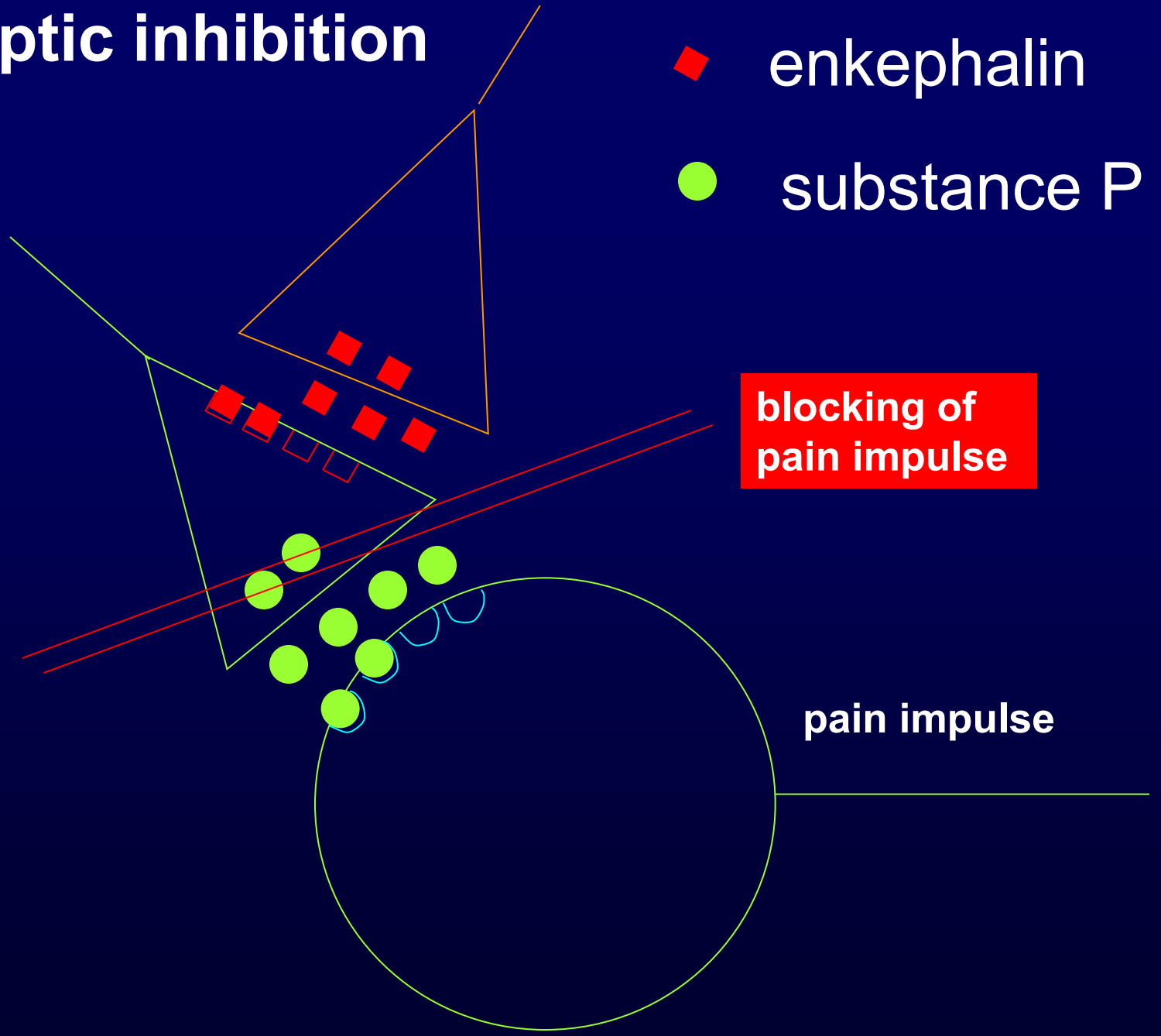
- in the subtantia gelatinosa
  - enkephalin secreting neuron is involved in presynaptic inhibition of the pain impulse transmission by blocking substance P release


substantia
gelatinosa
descending inhibitory tract
dorsal horn
c fibre input
substantia
gelatinosa cell

Presynaptic inhibition

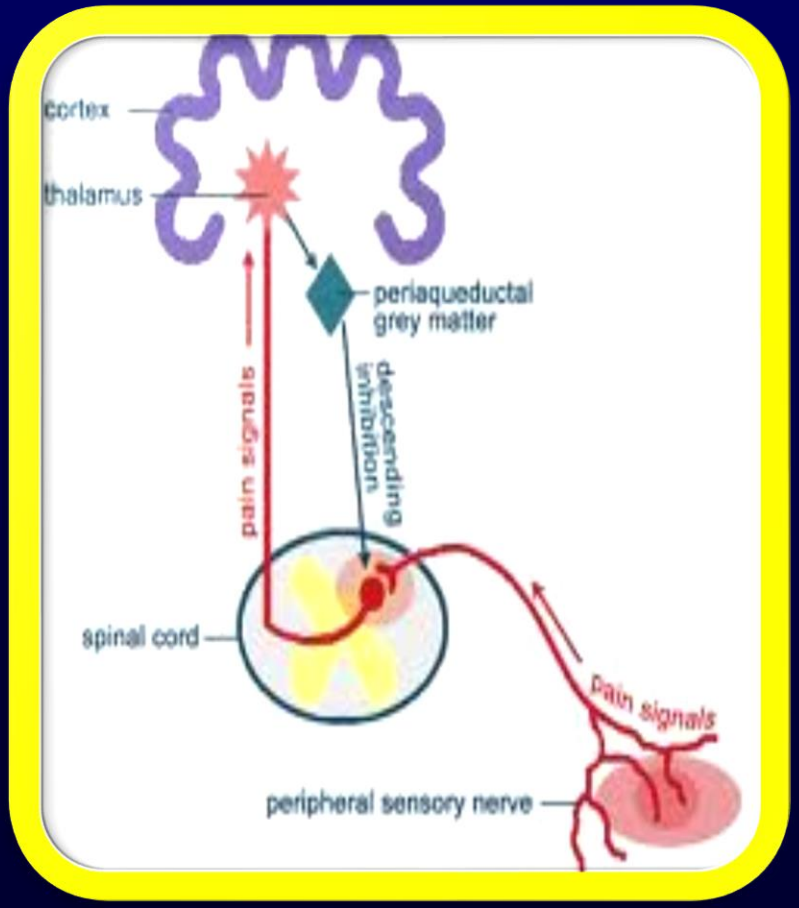


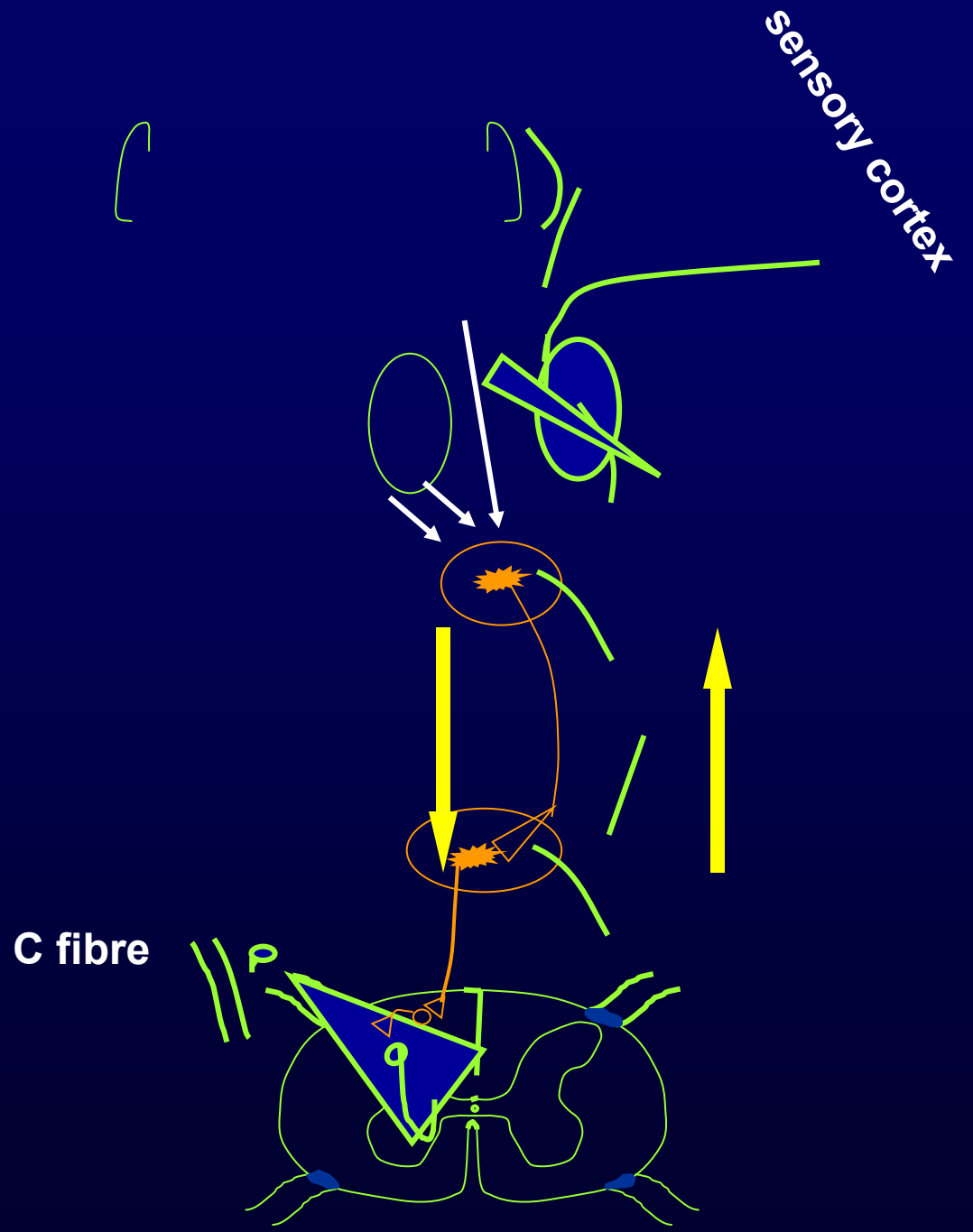
Presynaptic inhibition



PAIN PATHWAYS

- Think about pain perception and pain inhibition





Final pain perception depends on activity of the

- ❖Ascending pain impulse transmitting tracts
- ❖Descending pain modulatory (inhibitory) tracts

Cont.

- At SGR, there is a synapse between 1st order & 2nd order neuron. Also there is synapse of 2nd & 3rd order neuron at thalamus.
- The NT (neurotransmitter) at the synapse between Aδ fiber & 2nd order neuron at SGR is **glutamate** while the NT between C fiber & 2nd order neuron (slow pain) at SGR is **substance P.**

- Other 2nd order neurons, from SGR also are known, & they also cross to the opposite side to reach different areas of brain.

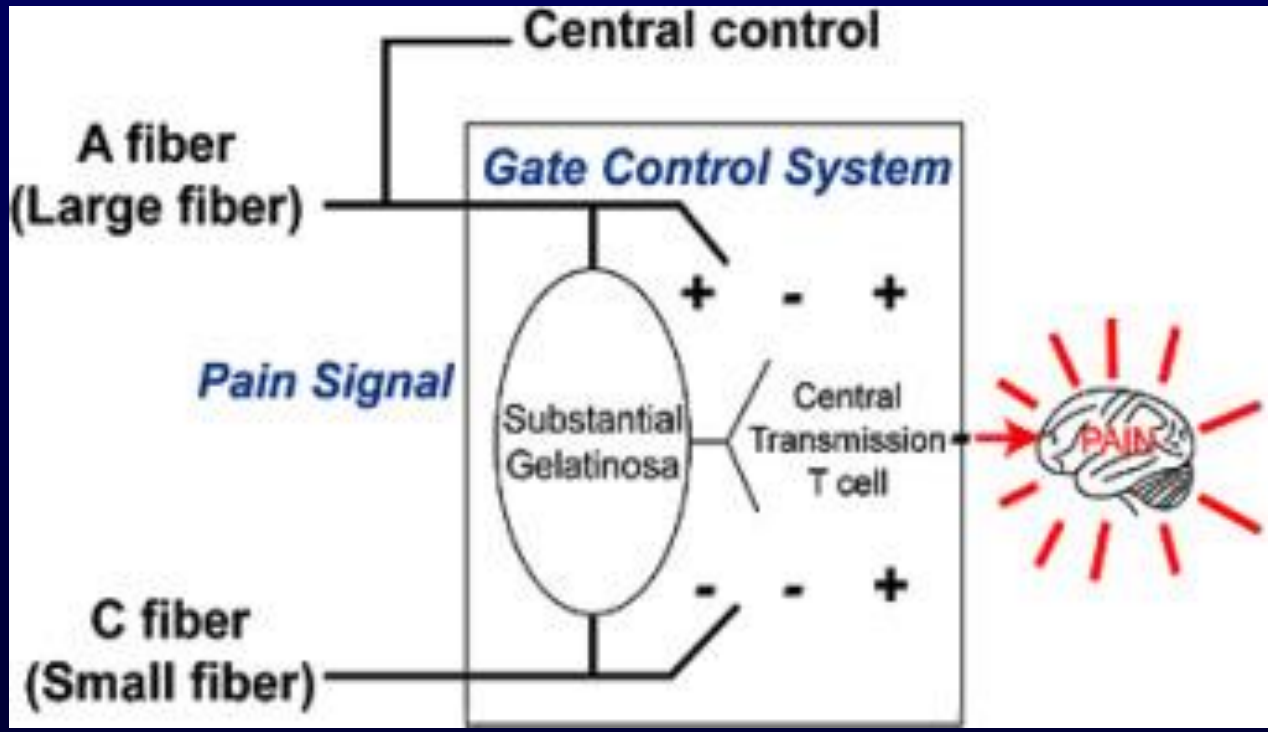
- The tip of dorsal horn, particularly the **SGR plays a key role in modification of pain perception.** It is called a **gate.** As a gate can be shut/ partially opened/ fully opened to control incoming traffic, so also behaves the SGR controlling the incoming traffic of pain.

Summary of receptor types

	Aß - fibres	Aδ - fibres	C- fibres
Threshold	Low	Medium	High
Axon diameter	6-14µm	1-6µm	0.2-1µm
Myelination	Yes	Thinly	No
Velocity	36-90	5-36	0.2-1
Receptor types	Mechanoreceptor	Mechano/Nociceptor	Nociceptor
Receptive field	Small	Small	Large
Quality	Touch	Sharp/first pain	Dull/second pain

Gate-Control Theory – Ronald Melzack (1960s)

- Described physiological mechanism by which psychological factors can affect the experience of pain.
- Neural gate can open and close thereby modulating pain.
- Gate is located in the spinal cord.
  - It is the SG

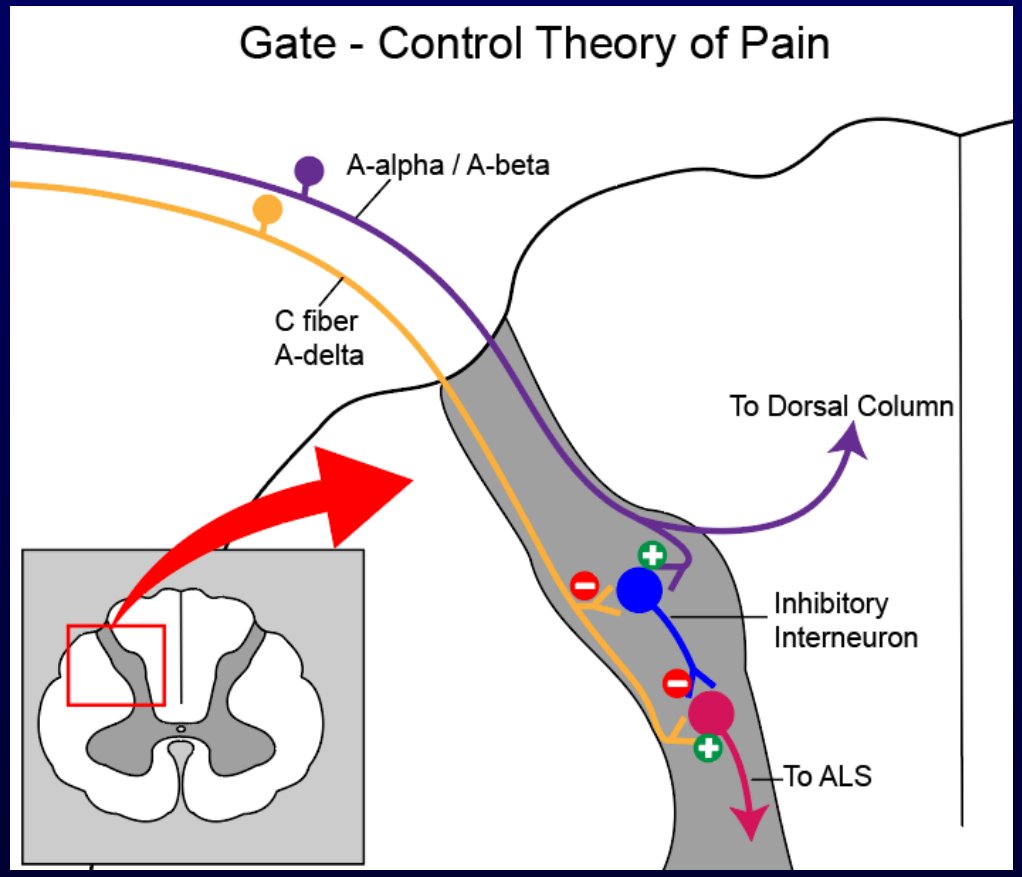


❖ Gate Control Theory

- **The large nerve fibers have collateral branches, which carry impulses to substantia gelatinosa where they stimulate secondary neurons.**
- The substantia gelatinosa cells terminate on the smaller nerve fibers just as the latter are about to synapse, thus reducing activity, the result is, ongoing activity is reduced or stopped –gate is closed.
- The theory also proposes that large diameter fiber input has ability to modulate synaptic transmission of small diameter fibers within the dorsal horn.
-

- Large diameter fibers transmit signals that are initiated by pressure, vibration and temperature; small diameter fibers transmit painful sensations.

- Activation of large fiber system inhibits small fiber synaptic transmission, which closes the gate to central progression of impulse carried by small fibers.





THANK YOU

Your collaboration and commitment make a meaningful difference—thank you.

